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Crime Pattern and Prediction UK

Student Name and SRN:

Muhammad Ruslan Babar | 24079307

Supervisor: Hariharan Muthu

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DECLARATION STATEMENT

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Student Name printed: Muhammad Ruslan Babar

Student Name signature:

Student SRN number: 24079307

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Abstract

This study analyses UK crime data to identify patterns and predict future crime levels using machine learning. The dataset from Police.uk was aggregated at LSOA-month level to capture spatial and temporal trends.

Exploratory Data Analysis showed that crime varies across time and location, with clear seasonal patterns and geographic hotspots. Clustering methods were used to detect high-risk areas, where DBSCAN performed better than KMeans in identifying dense clusters and outliers.

For prediction, Linear Regression and Random Forest were applied, with Random Forest achieving better performance. After tuning, the final model reached an RMSE of 4.62. Feature analysis showed that past crime (lag1) is the most important predictor.

Overall, the study demonstrates that combining data analysis, clustering, and machine learning is effective for understanding and predicting crime patterns.

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1. Introduction

1.1. Background

Crime is a major issue that affects public safety, economic activity, and quality of life. In the United Kingdom, crime data is openly available through platforms such as Police.uk, allowing analysis using data-driven methods. Crime data is large and complex, including information on location, type, outcome, and time. It is also unevenly distributed, with certain areas experiencing higher crime levels, known as crime hotspots.

1.2. Motivation

Due to the size and complexity of crime data, identifying patterns manually is difficult. Traditional methods often fail to capture hidden spatial and temporal relationships, creating a need for more advanced analytical approaches.

1.3. Data Analysis and Machine Learning

Exploratory Data Analysis (EDA) helps identify trends and patterns in the data, such as changes over time and geographic concentration (Brownlee, 2020). Machine learning techniques extend this by enabling prediction and pattern detection. Clustering methods such as KMeans and DBSCAN can identify hotspots, while regression models like Linear Regression and Random Forest can predict future crime levels (Khan et al., 2019).

1.4. Model Interpretation

Understanding model decisions is important. Techniques such as SHAP help explain how different features influence predictions, improving transparency (Lundberg and Lee, 2017).

1.5. Research Gap

Many studies focus on either prediction or clustering alone. There is a need for a combined approach that includes exploration, hotspot detection, prediction, and evaluation.

1.6. Aim and Objectives

The aim of this study is to analyse crime data and predict future crime levels using machine learning.

Objectives include:

- Analysing spatial and temporal patterns
- Identifying crime hotspots
- Building and evaluating predictive models

1.7. Research Questions

- How is crime distributed across areas and time?
- Which areas are high-risk hotspots?

- How accurately can crime be predicted?

1.8. Scope of Study

This study uses UK police crime data aggregated at LSOA and monthly levels. It includes preprocessing, analysis, clustering, prediction, and evaluation, but not real-time deployment.

1.9. Workflow Overview

The overall workflow is shown in Figure 2.1.1, illustrating the process from data collection to model evaluation.

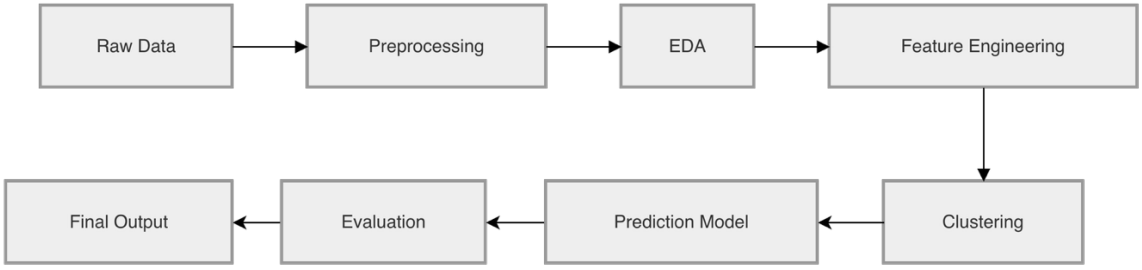

Figure 1.1: Overall workflow of crime data analysis and prediction

2. Literature Review

2.1. Background

Crime analysis has evolved from simple statistical methods to more advanced machine learning approaches. Traditional techniques focused on descriptive statistics and basic visualisations, which were limited in handling large and complex datasets.

Recent research shows that crime is influenced by both spatial and temporal factors. Crime levels vary across locations and time due to factors such as population density and urban structure. Identifying these patterns is important for detecting high-risk areas and supporting effective decision-making (Chainey and Ratcliffe, 2013).

Exploratory Data Analysis (EDA) is commonly used to understand crime data. It helps identify trends, patterns, and anomalies using visual tools such as line charts, bar graphs, and heatmaps (Brownlee, 2020).

2.2. Crime Hotspot Detection

Hotspot detection is a key task in crime analysis, as it identifies areas with high crime concentration.

Clustering methods are widely used for this purpose. KMeans groups data into a fixed number of clusters based on similarity, but it requires the number of clusters to be predefined and may struggle with irregular patterns. In contrast, DBSCAN groups data based on density and does not require a predefined number of clusters. It is more effective in detecting real-

world hotspots and can also identify outliers, making it suitable for identifying extreme crime areas.

2.3. Crime Prediction Using Machine Learning

Machine learning models are commonly used to predict future crime levels. Linear Regression is a simple baseline model that assumes a linear relationship between variables but may not perform well with complex data.

Random Forest is a more advanced model that captures non-linear relationships and interactions between features. It generally provides better prediction accuracy and has been widely used in crime prediction studies.

Model performance can be further improved using techniques such as hyperparameter tuning. Methods like Randomised Search help identify the best model settings, reducing prediction error.

2.4. Feature Importance and Model Interpretation

Understanding model predictions is important in crime analysis. SHAP is a widely used method that explains how each feature contributes to predictions. For example, past crime levels and location features often have a strong influence on results.

This improves transparency and ensures that predictions are based on meaningful factors.

2.5. Research Gap

Many studies focus only on a single aspect, such as clustering or prediction. Few combine multiple techniques into one framework. In addition, model interpretation is often overlooked.

This study addresses these gaps by integrating exploratory analysis, clustering (KMeans and DBSCAN), prediction models (Linear Regression and Random Forest), and interpretation methods (SHAP) into a single workflow.

3. Methodology

3.1. Overview

This study follows a structured data-driven approach to analyse crime patterns and predict future crime levels. The process begins with data collection and preparation, followed by exploratory analysis to understand trends and relationships within the dataset. Clustering techniques are then applied to identify crime hotspots, and machine learning models are used to predict future crime levels.

Each step in this process is important because the quality of the final results depends on earlier stages. For example, proper data cleaning improves model performance, while meaningful feature selection helps the model focus on important variables (Brownlee, 2020).

3.2. Data Collection

The dataset used in this study was obtained from the UK Police open crime data platform (UK Government, 2023). It contains detailed records of reported crimes, including variables such as crime type, location, date, and outcome category.

The original dataset is recorded at street level, which can be noisy and difficult to analyse. To improve usability, the data was aggregated at LSOA-month level, meaning that crimes were grouped by area and by month. This approach reduces noise and allows clearer identification of patterns over time and across locations.

Figure 3.2 shows a sample of the aggregated dataset, where each row represents the total number of crimes in a specific area and month.

	LSOA code	year	monthNum	crimeCount
0	E01000001	2023	10	6
1	E01000001	2023	11	19
2	E01000001	2023	12	6
3	E01000001	2024	1	14
4	E01000001	2024	2	8

Figure 3.1: Aggregated dataset at LSOA-month level

3.3. Data Preprocessing

Data preprocessing was carried out to clean and prepare the dataset before analysis. Real-world data often contains missing values and inconsistencies, which can affect model performance.

Missing values in key variables such as latitude, longitude, and LSOA code were removed to ensure accuracy in spatial analysis. Data types were also checked and corrected, particularly for time-related fields.

A new variable called crimeCount was created to represent the total number of crimes per LSOA and month. This serves as the target variable for prediction.

To improve data distribution, a logarithmic transformation (logCrime) was applied. This helps reduce skewness and makes the data more suitable for modelling (Osborne, 2010).

Additional features were created to capture important patterns:

- **monthNum and year** for temporal trends
- **latitude and longitude** for spatial patterns

These features allow the model to better understand how crime varies across time and location.

3.4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the dataset before applying machine learning models. This step helps identify key trends, patterns, and relationships within the data, which are important for selecting features and building accurate models (Brownlee, 2020).

The monthly crime trend was analysed to observe how crime levels change over time. Figure 3.3 shows that crime varies across different months rather than remaining constant. This indicates that crime activity is influenced by time and may follow certain patterns over the study period.

Figure 3.2: Monthly crime trend

Seasonal patterns were further examined by grouping the data by month number.
Figure 3.4 shows that some months consistently experience higher or lower crime levels. This suggests the presence of seasonal effects, which may be linked to factors such as weather, public activity, or social behaviour.

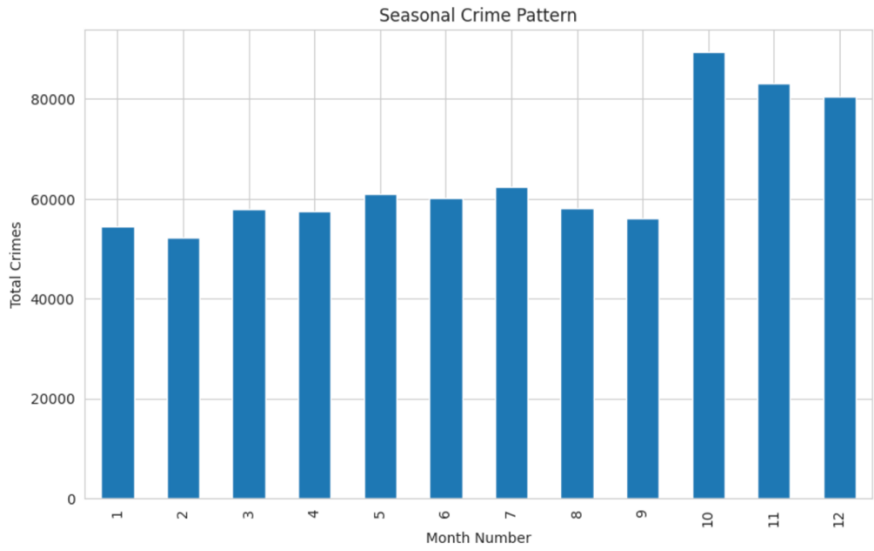

Figure 3.3: Seasonal crime pattern

Bar charts were used to analyse the distribution of crime types.

Figure 3.5 shows that certain crimes, such as theft and violence, occur more frequently than others. This highlights the dominant types of crime in the dataset and helps in understanding the overall composition of criminal activity.

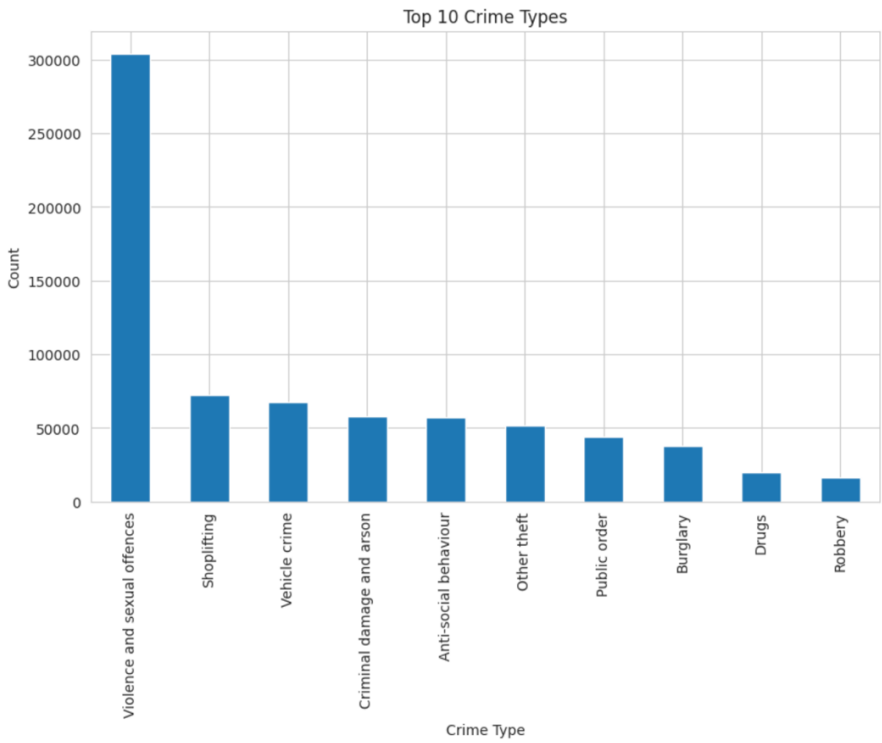

Figure 3.4: Top crime types

Spatial analysis was conducted using latitude and longitude values to examine how crime is distributed across locations.

Figure 3.6 shows that crime is concentrated in specific geographic areas rather than being evenly spread. These high-density regions indicate potential crime hotspots, which are important for further analysis and clustering.

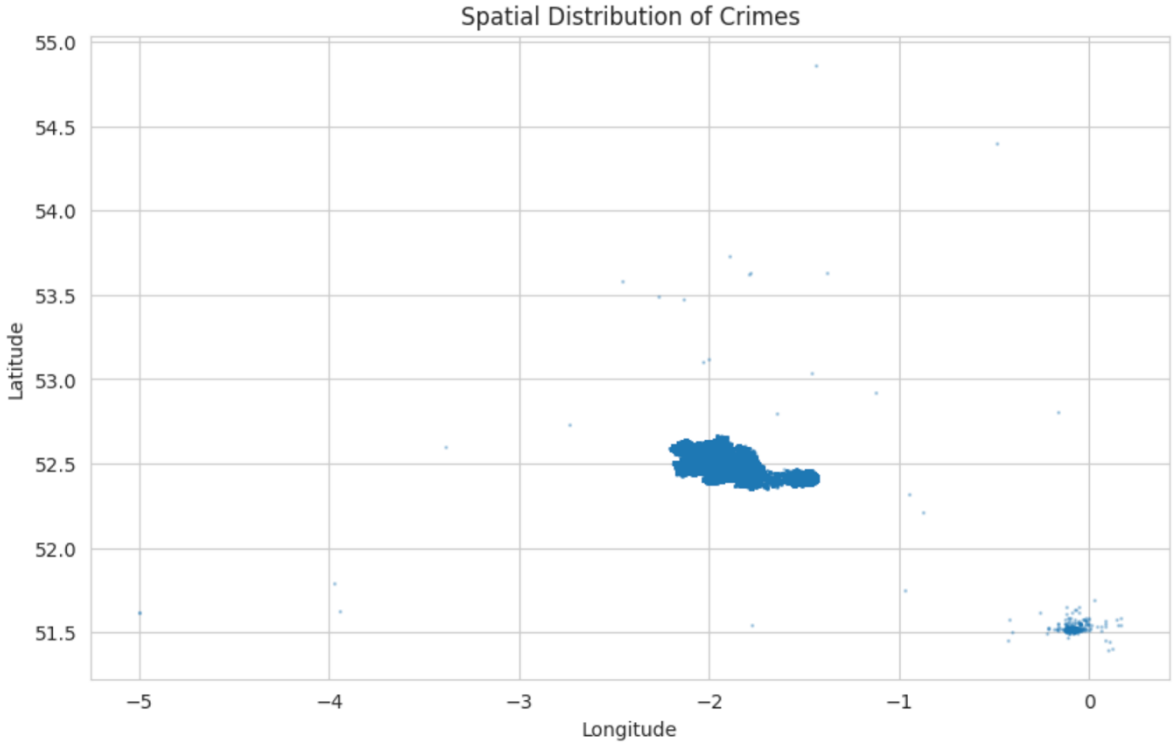

Figure 3.5: Spatial distribution of crime

Overall, the EDA results show that crime is influenced by both temporal and spatial factors. These findings support the use of time-based and location-based features in the modelling process and provide a strong foundation for subsequent clustering and prediction tasks.

3.5. Feature Engineering and Selection

Feature engineering was used to create variables that improve model performance.

A key feature is **lag1**, which represents the crime count from the previous month. This is important because crime often follows temporal patterns, where past values influence future outcomes.

Another important feature is **dominantTypeCount**, which captures the most common crime type in a given area. This helps represent the nature of crime in each location.

A seasonal feature **isWinter** was also introduced to capture potential seasonal effects.

Feature selection was performed using statistical methods such as SelectKBest (Guyon and Elisseeff, 2003). The results showed that lag1 and dominantTypeCount are the most influential features, while variables such as year have limited impact.

Figure 3.7 shows the correlation between features and highlights strong relationships with the target variable.

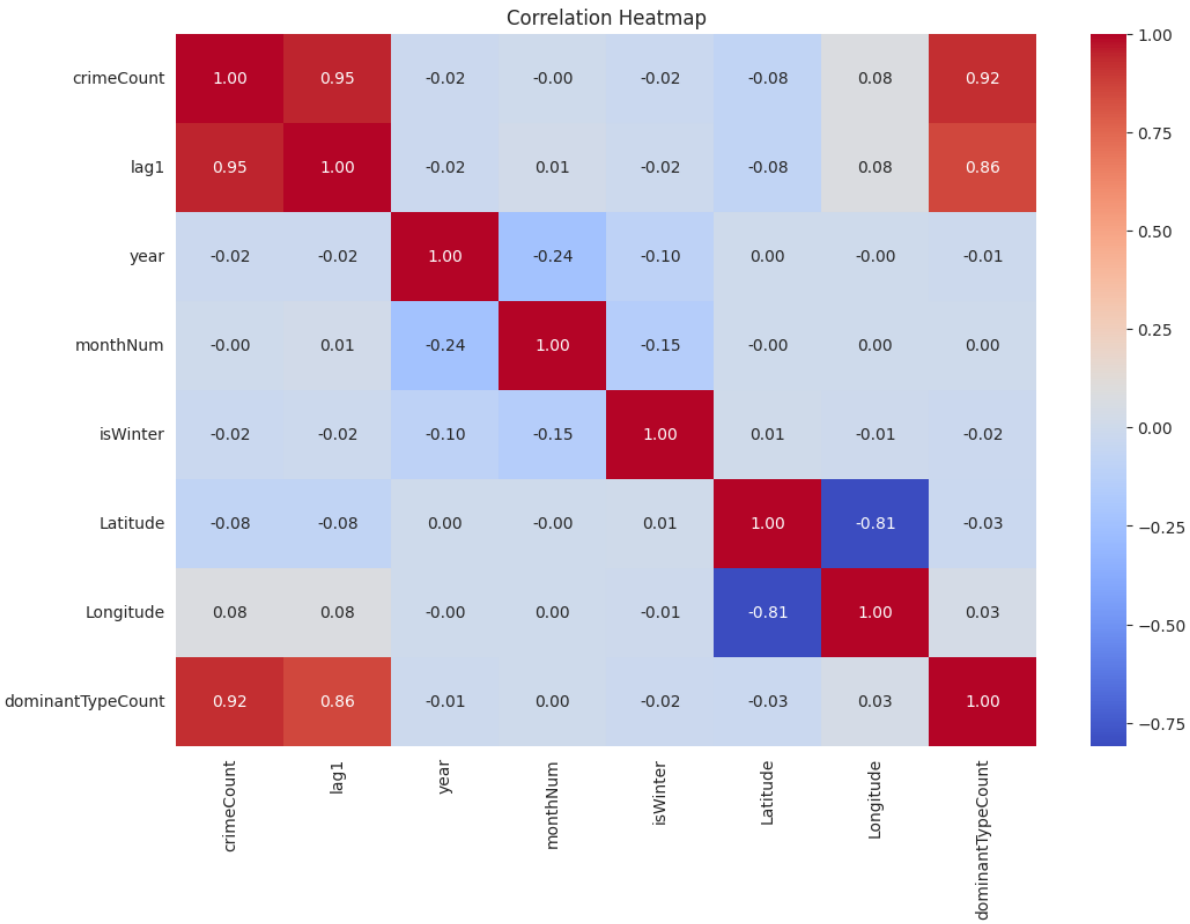

Figure 3.6: Feature correlation heatmap

3.6. Clustering for Hotspot Detection

Clustering techniques were applied to identify areas with similar crime patterns and detect hotspots.

3.6.1. KMeans Clustering

KMeans groups data into a fixed number of clusters based on distance. It assigns each data point to the nearest cluster centre.

Figure 3.8 shows that KMeans forms clear clusters across the dataset. However, it does not clearly separate extreme high-crime areas, as it assumes clusters of similar size and shape.

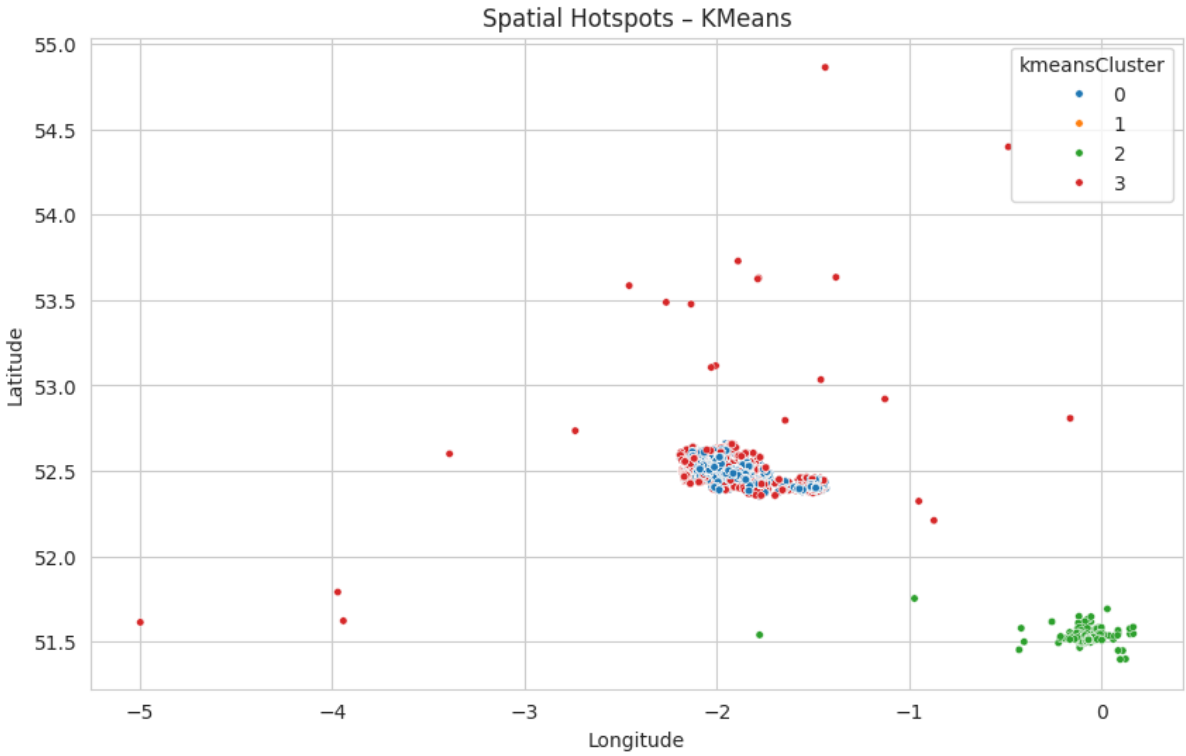

Figure 3.7: KMeans clustering results

3.6.2. DBSCAN Clustering

DBSCAN groups data based on density rather than distance. It identifies clusters where data points are densely packed and labels sparse points as noise.

Figure 3.9 shows that DBSCAN is able to identify dense clusters representing high-crime areas. It also identifies outliers, which are areas with very low or unusual crime levels.

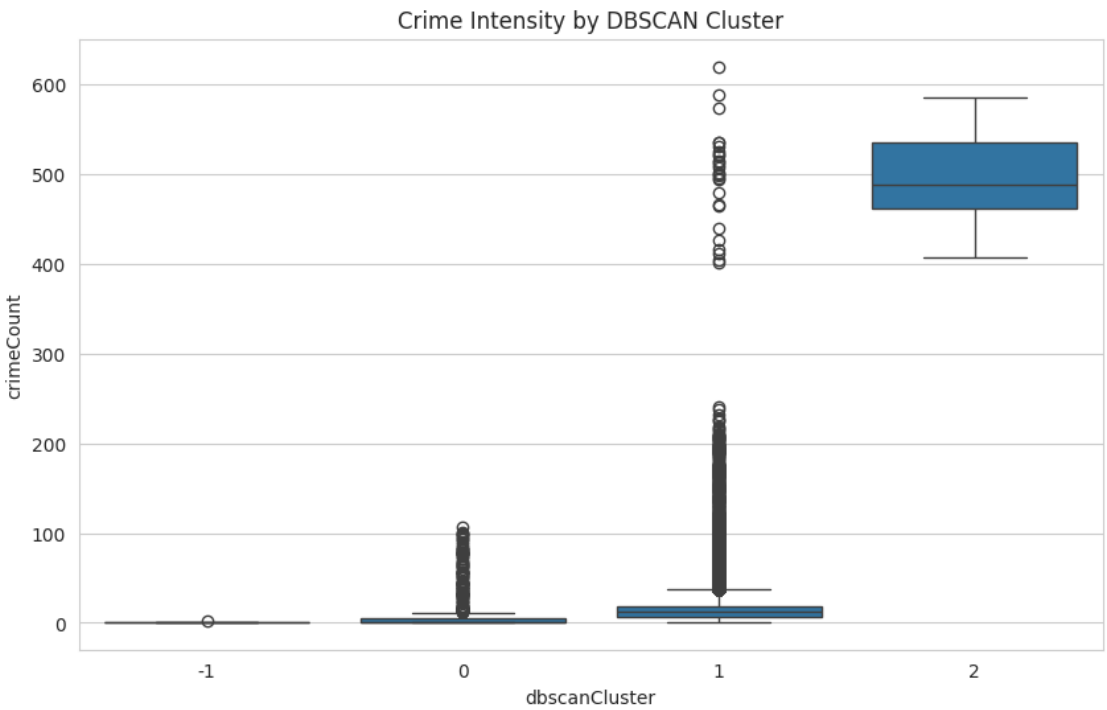

Figure 3.8: DBSCAN clustering results

Results Comparison

The results show clear differences between the two methods:

- KMeans provides general grouping but may miss extreme patterns
- DBSCAN more effectively identifies **true crime hotspots and high-density regions**
- DBSCAN also detects **outliers**, which can represent unusual or isolated areas

Overall, DBSCAN performs better for hotspot detection because crime data is naturally uneven and clustered.

3.7. Prediction Modelling

To predict future crime levels, two machine learning models were used: Linear Regression and Random Forest.

Linear Regression was used as a baseline model. It assumes a simple linear relationship between features and the target variable (Montgomery et al., 2012). While it is easy to understand, it cannot capture complex patterns in the data.

Random Forest was used as the main model because it can handle non-linear relationships. It works by combining multiple decision trees to improve prediction accuracy (Breiman, 2001). This makes it more suitable for complex datasets like crime data.

3.8. Model Training and Evaluation

The dataset was split into training and testing sets based on time. Earlier data was used for training, while later data was used for testing.

Model performance was evaluated using RMSE, which measures prediction error (Hyndman and Koehler, 2006).

The initial Random Forest model showed signs of overfitting, with lower training error compared to testing error. To address this, regularisation techniques were applied by limiting tree depth and sample size.

Further improvement was achieved using Randomised Search for hyperparameter tuning. This resulted in a lower RMSE and improved generalisation.

Figure 3.10 shows actual vs predicted values, where points close to the line indicate accurate predictions.

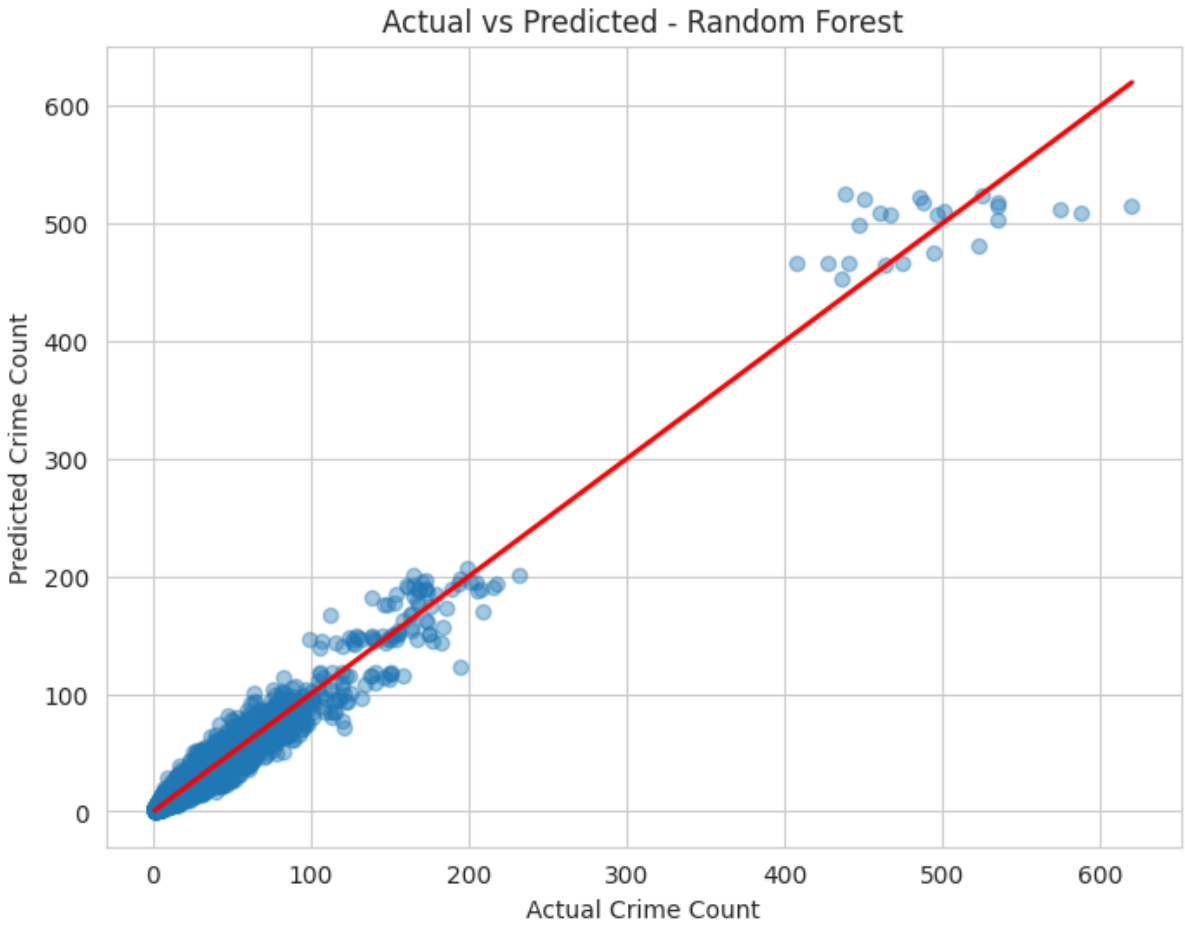

Figure 3.9: Actual vs predicted values

Figure 3.11 shows the residual distribution, where most errors are close to zero, indicating good model performance.

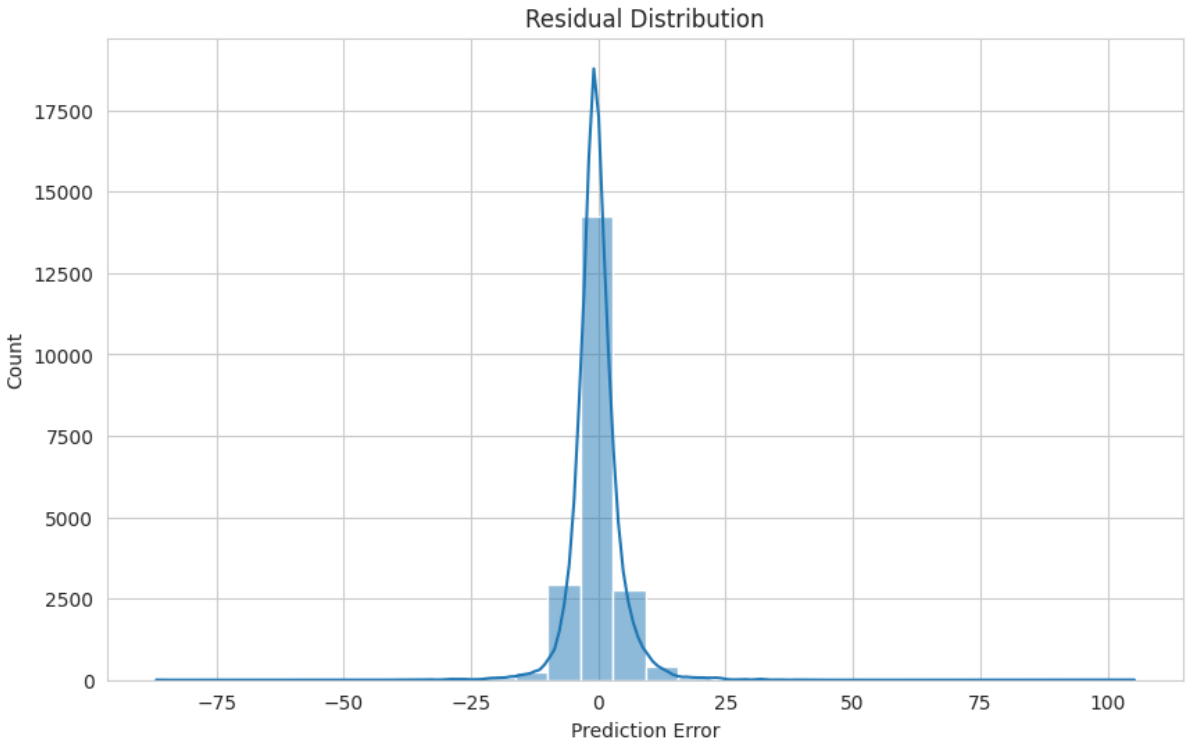

Figure 3.10: Residual distribution

3.9. Model Interpretation

To understand model behaviour, SHAP analysis was used (Lundberg and Lee, 2017).

Figure 3.12 shows that:

- lag1 has the highest impact on predictions
- dominantTypeCount is also important
- Location features contribute less but still provide useful information

This confirms that past crime is a strong predictor of future crime.

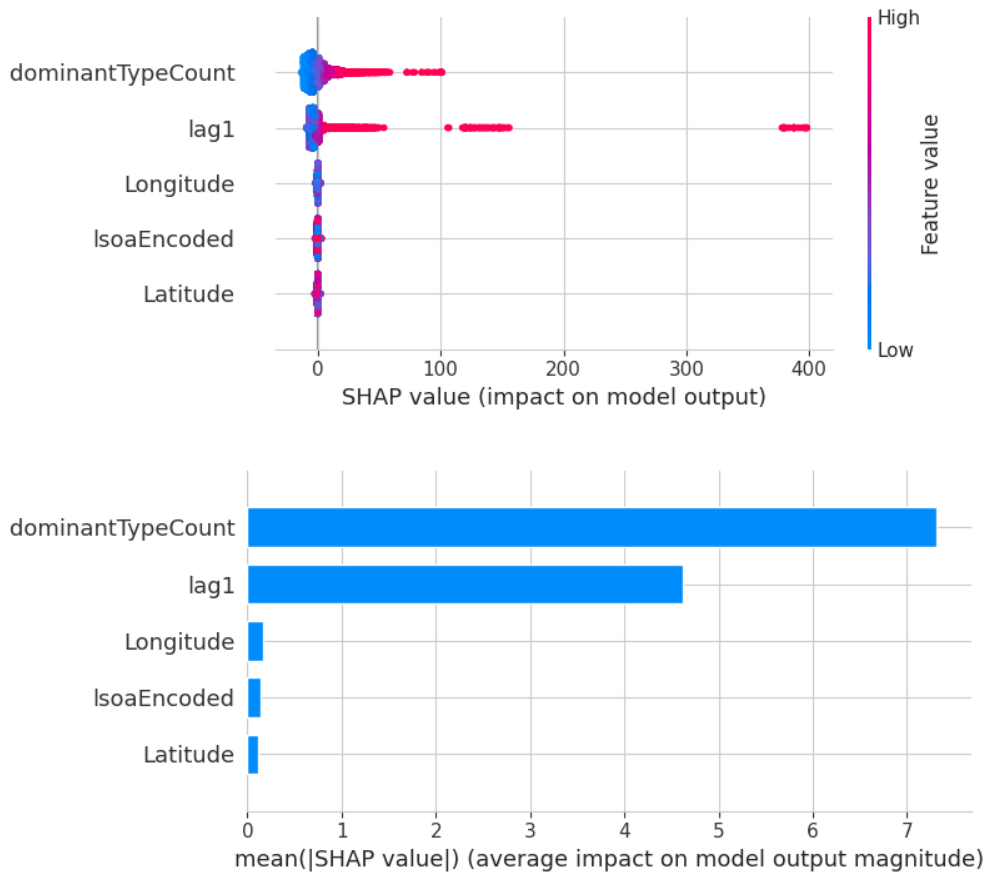

Figure 3.11: SHAP feature importance

3.10. Prediction Function

A function called predictNextMonthCrime was developed to apply the model in practice.

This function takes input values such as location and previous crime levels, and returns a predicted crime count for the next month. It also assigns a risk level (Low, Medium, High, Very High) based on the predicted value.

This makes the model easier to use for decision-making and real-world applications.

4. Results And Analysis

4.1. Overview

This chapter presents and interprets the results obtained from the analysis and modelling process. Unlike the previous chapter, which focused on the methodology, this section focuses on explaining the meaning of the results and their implications.

The findings are discussed in a structured way, starting from exploratory insights, followed by feature relationships, clustering results, prediction performance, and model validation. The goal is to understand how crime patterns behave and how well the models capture these patterns.

4.2. Exploratory Data Analysis Results

Exploratory Data Analysis provides important insights into how crime behaves across time, categories, and locations.

The monthly crime trend shows noticeable fluctuations over time.

Figure 4.1 shows that crime levels rise and fall rather than following a stable pattern. This suggests that crime is influenced by dynamic factors such as social activity, policing, or economic conditions.

Figure 4.1: Monthly crime trend

The seasonal analysis provides further insight into repeating patterns.

Figure 4.2 shows that certain months consistently experience higher crime levels. This confirms that crime follows seasonal behaviour, which may be linked to weather conditions or increased public activity during specific periods.

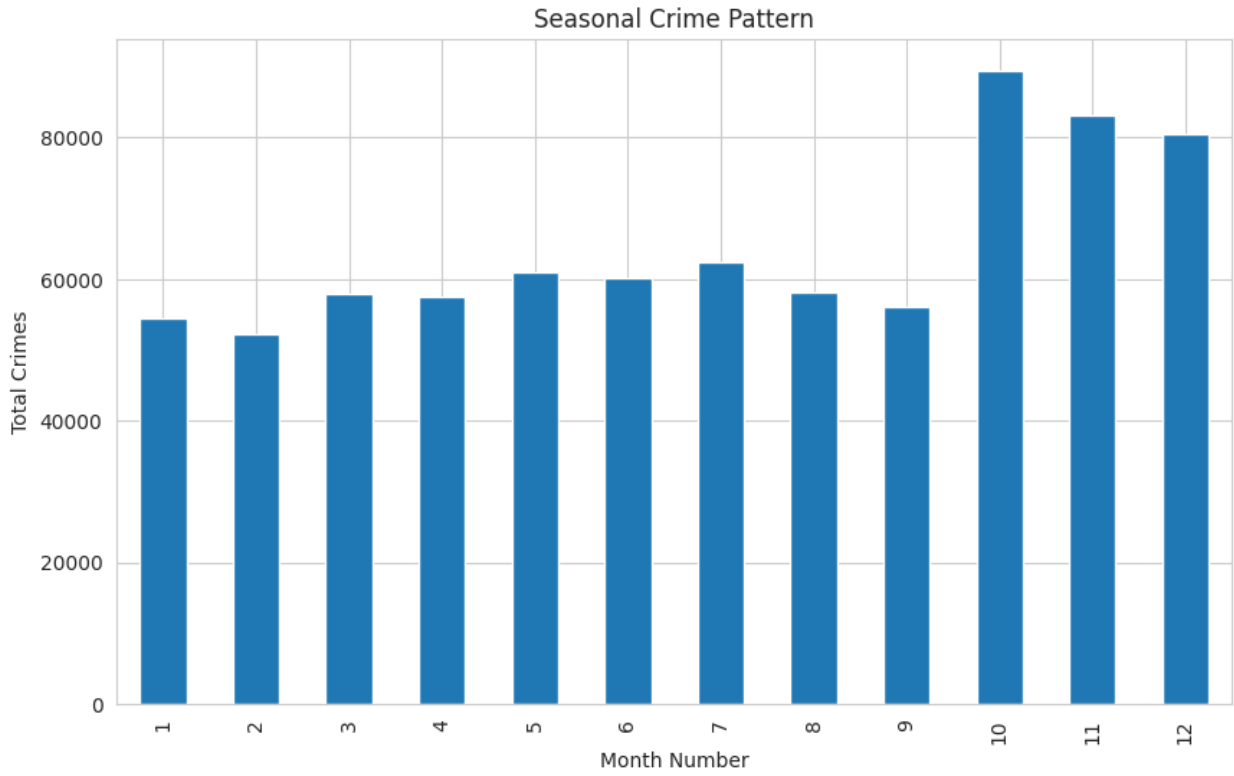

Figure 4.2: Seasonal crime pattern

The distribution of crime types highlights the composition of crime in the dataset.

Figure 4.3 shows that a small number of crime types dominate, such as theft and violence. This indicates that crime is not evenly distributed across categories, and certain types contribute more heavily to overall crime levels.

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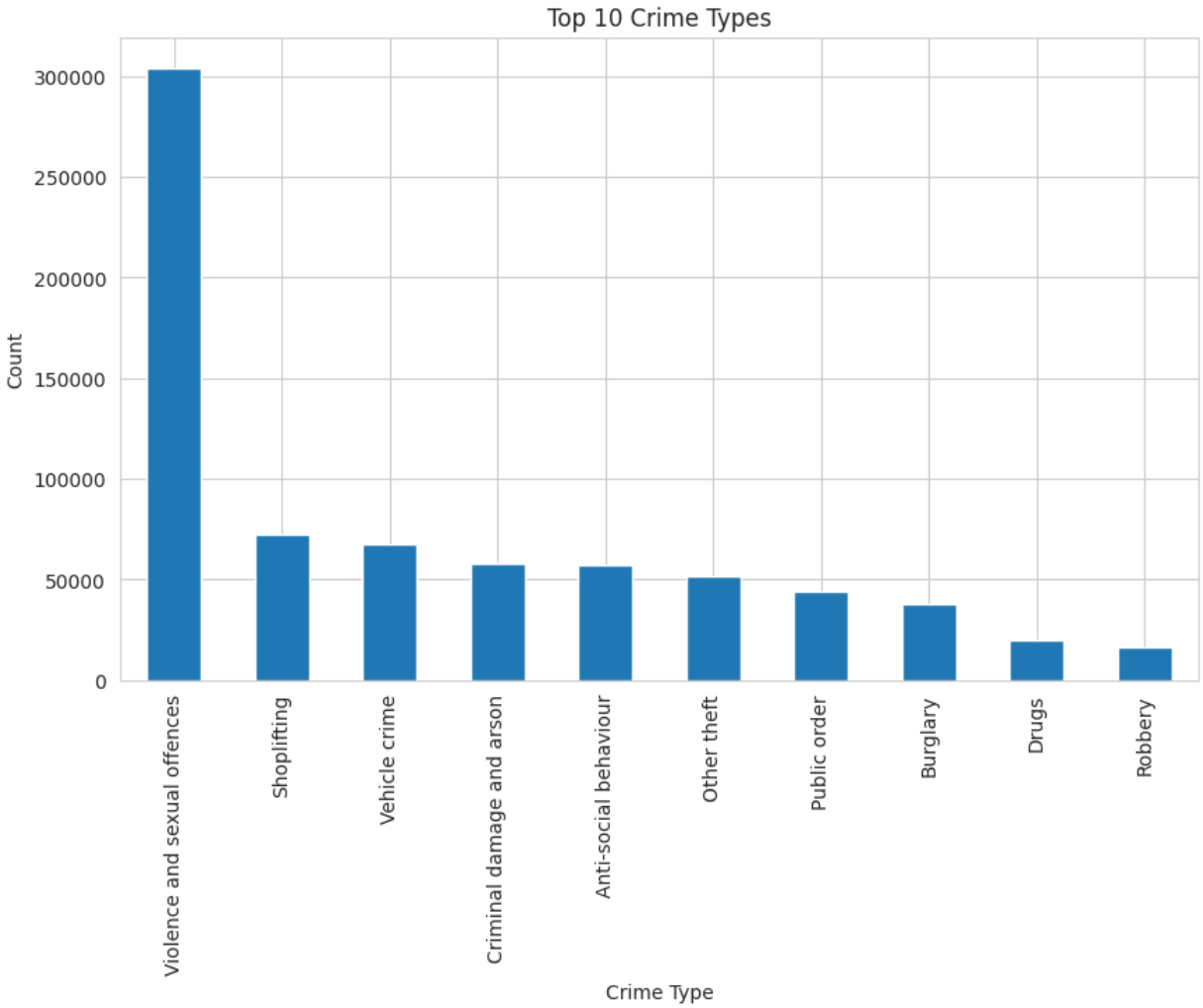

Figure 4.3: Top crime types

Spatial analysis clearly reveals geographic concentration.

Figure 4.4 shows clusters of high-density crime areas, confirming the existence of hotspots. This suggests that location plays a critical role in crime patterns and must be considered in further analysis.

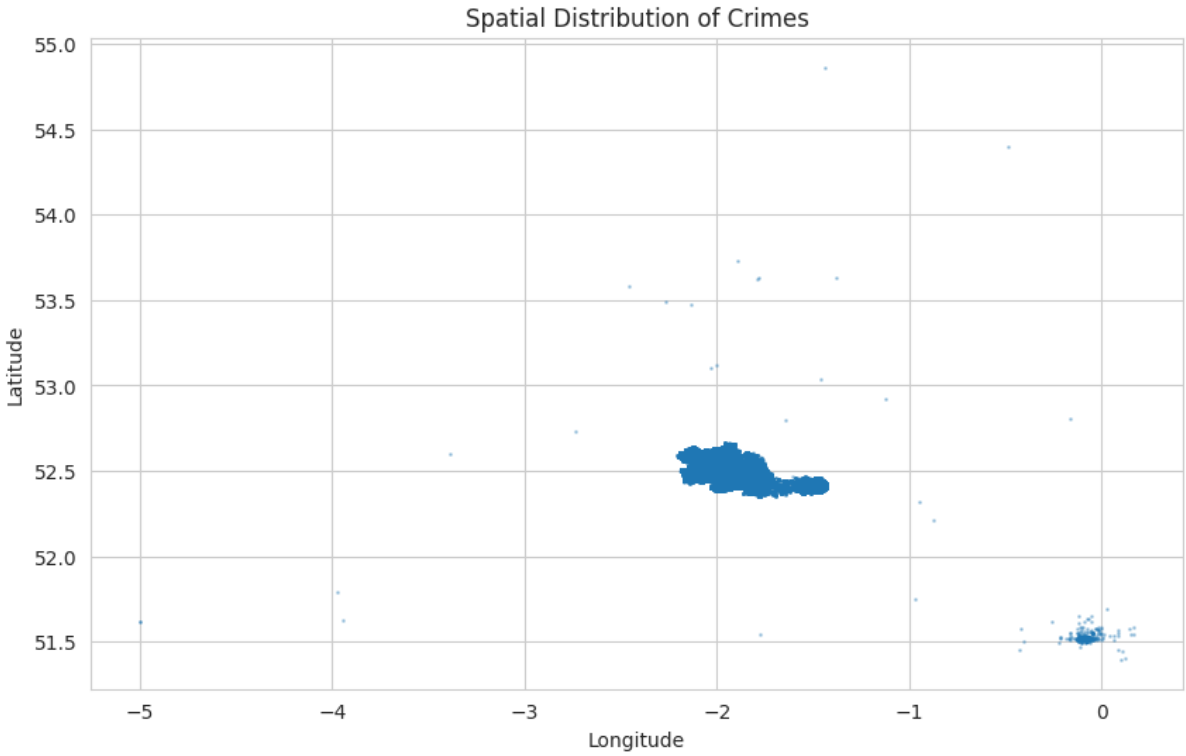

Figure 4.4: Spatial distribution of crime

Overall, these findings confirm that crime is strongly influenced by both time and location, supporting the modelling approach used in this study.

4.3. Feature Relationships and Correlation Analysis

Understanding relationships between features helps explain why the model makes certain predictions.

Figure 4.5 shows the correlation heatmap, which highlights strong relationships between variables.

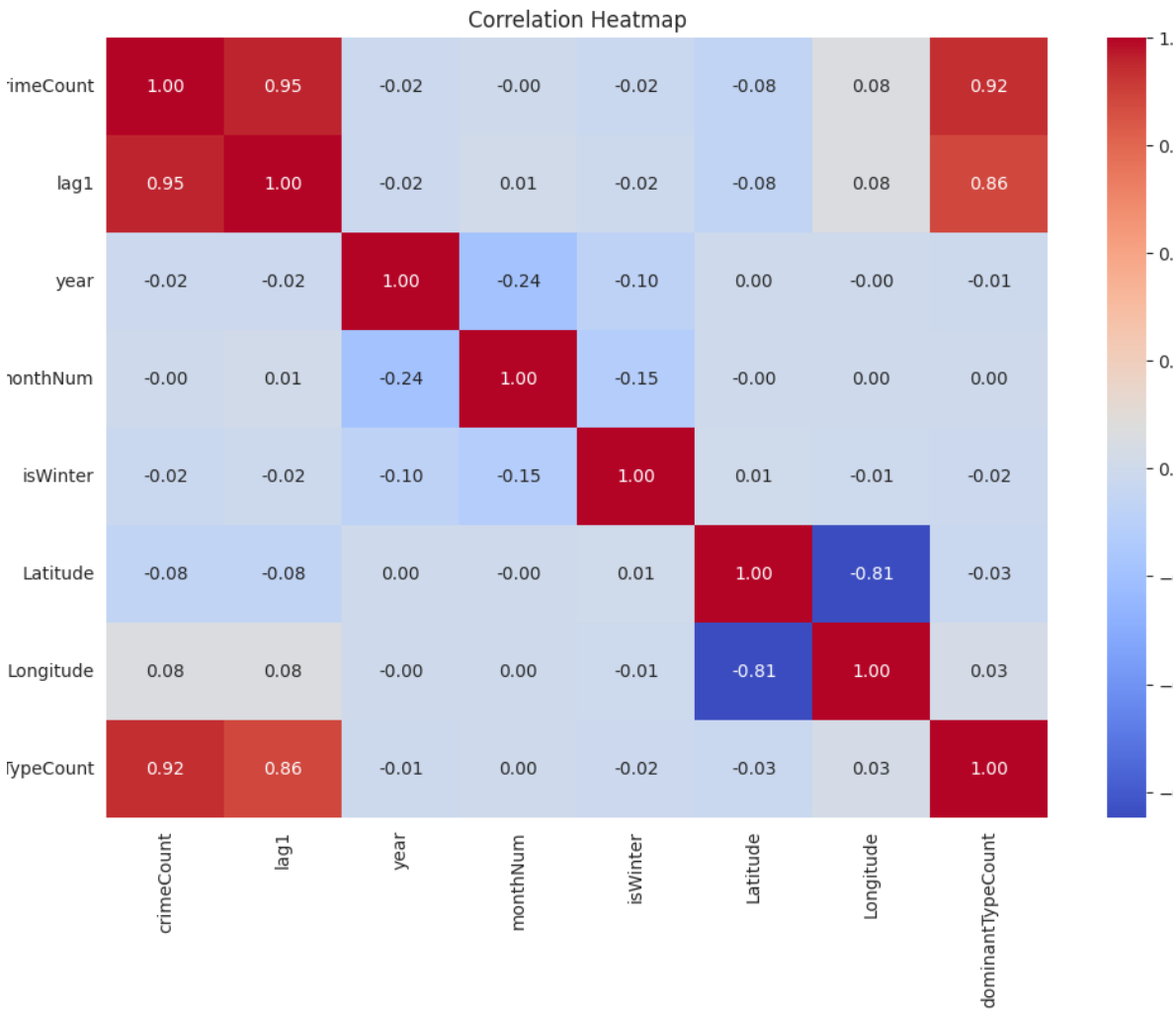

Figure 4.5: Feature correlation heatmap

The most important finding is the strong positive correlation between lag1 and crime count. This indicates that past crime levels are a strong predictor of future crime. Such temporal dependency is common in crime data, where behaviour tends to persist over time.

Similarly, dominantTypeCount shows a strong relationship with crime count. This suggests that areas with frequent occurrence of certain crime types tend to have higher overall crime levels.

Other variables, such as year and monthNum, show weak direct correlation. This suggests that their influence is more indirect or captured through other features.

Spatial features such as latitude and longitude show low correlation values, but this does not reduce their importance. Instead, their value lies in identifying geographic patterns rather than direct prediction.

These findings confirm that temporal and crime-type features are the strongest predictors in the model.

4.4. Clustering Results and Hotspot Identification

Clustering was used to identify crime patterns across different regions and detect hotspots.

4.4.1. KMeans Clustering Analysis

Figure 4.6 shows the clusters formed using KMeans.

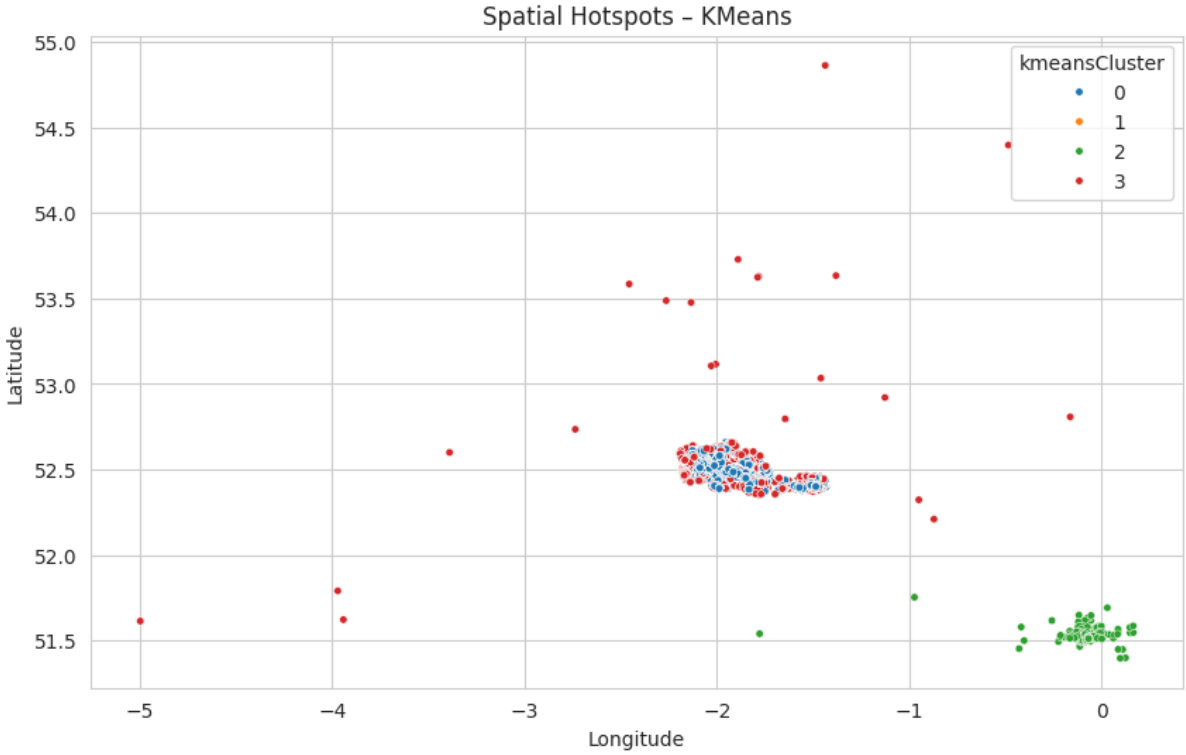

Figure 4.6: KMeans clustering results

The results indicate that KMeans is able to group data into clusters, but the separation between clusters is not very strong. The relatively low silhouette score suggests that the clusters overlap and are not well-defined.

This happens because KMeans assumes clusters are evenly shaped and distributed. However, crime data is irregular and uneven, which makes KMeans less effective in capturing real-world patterns.

As a result, KMeans provides only a general grouping and does not clearly isolate high-crime hotspots.

4.4.2. DBSCAN Clustering Analysis

DBSCAN provides a more realistic representation of crime patterns.

Figure 4.7 shows the distribution of crime intensity across DBSCAN clusters..

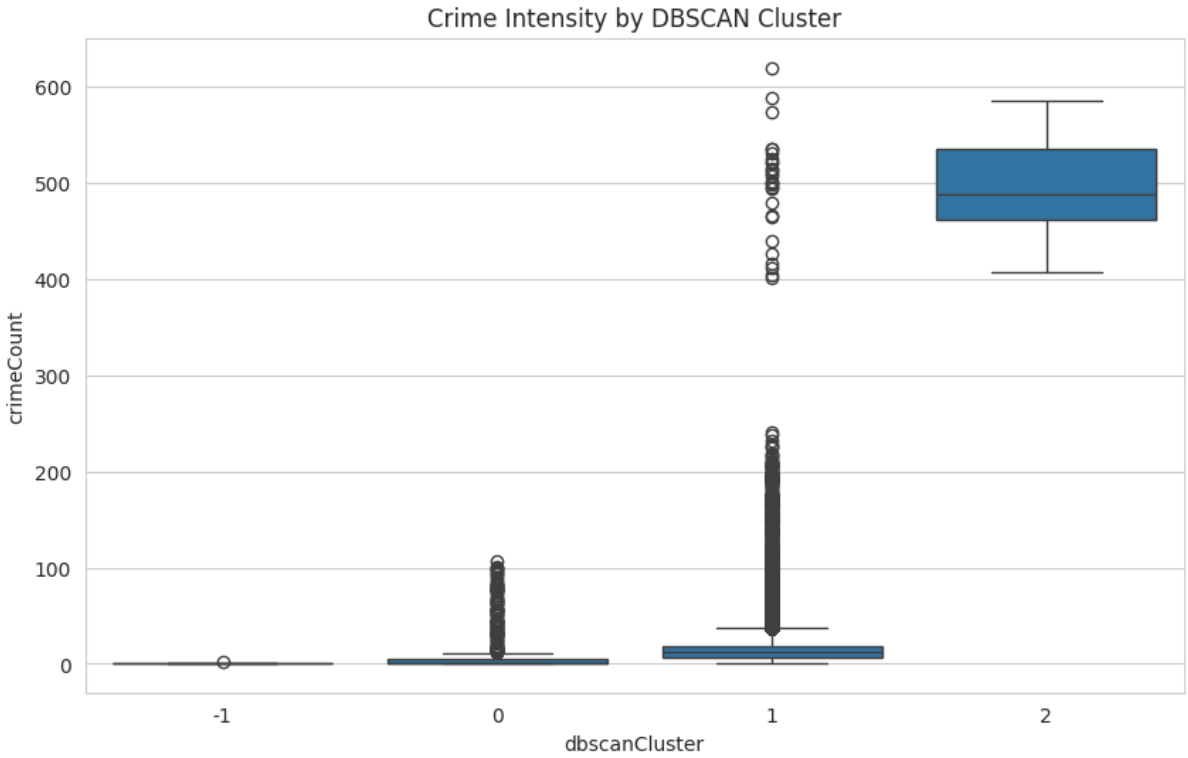

Figure 4.7: Crime intensity by DBSCAN cluster

The results clearly identify clusters with different crime levels:

- Cluster 2 contains very high crime values (hotspots)
- Cluster 1 represents moderate crime areas
- Cluster 0 represents lower crime regions
- Label -1 represents noise or outliers

The boxplot shows that cluster 2 has significantly higher crime values compared to others. This confirms that DBSCAN successfully isolates extreme high-crime areas.

Outliers (label -1) represent unusual or low-density areas. These points are important because they highlight regions that do not follow common patterns.

Overall, DBSCAN performs better because it captures density-based patterns, making it more suitable for real-world crime data.

4.5. Prediction Model Performance

Two models were compared to evaluate prediction accuracy.

Figure 4.8 shows the model performance comparison.

	Model	RMSE
0	Linear Regression	5.354560
1	Random Forest	4.843927
2	Regularised RF	4.859432
3	Tuned RF	4.626201

Figure 4.8: Model performance comparison

Linear Regression performed as expected for a simple model but showed higher error. This is because it assumes a linear relationship, which does not match the complexity of crime data.

Random Forest achieved lower RMSE, indicating better performance. This shows that it is able to capture non-linear relationships and interactions between features.

The results confirm that advanced models are more suitable for complex datasets like crime data.

4.6. Overfitting and Model Improvement

The initial Random Forest model showed signs of overfitting.

Figure 4.9 shows that training error is lower than testing error, meaning the model learned the training data too closely but struggled on new data.

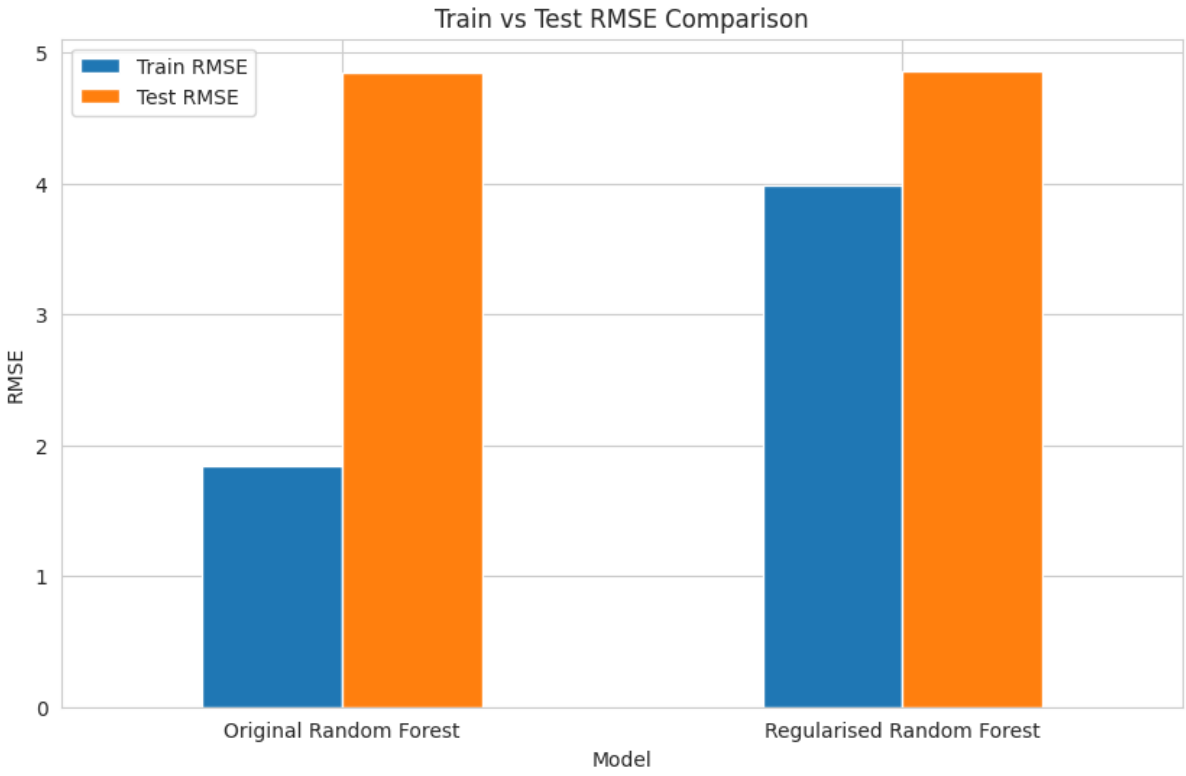

Figure 4.9: Train vs Test RMSE

To fix this, regularisation techniques were applied:

- Limiting tree depth
- Increasing minimum samples per split

After these changes, the gap between training and testing error reduced, showing improved generalisation.

4.7. Feature Selection and Impact

Feature selection results are shown in Figure 4.10.

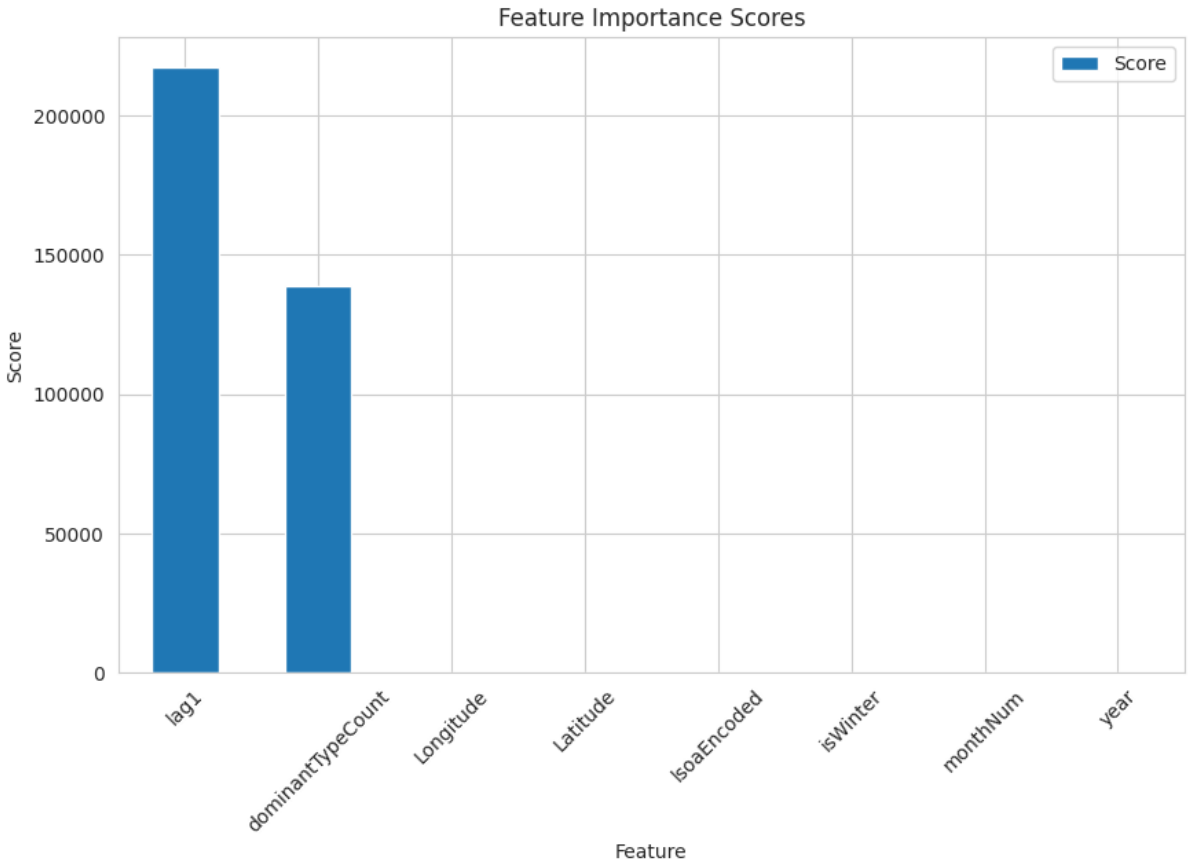

Figure 4.10: Feature importance scores

The most important feature is lag1, followed by dominantTypeCount. This confirms that past crime behaviour is the strongest predictor of future crime.

Less important features were removed without affecting performance, which simplified the model and improved efficiency.

4.8. SHAP Analysis and Model Interpretation

SHAP analysis provides deeper insight into how the model makes decisions.

Figure 4.11 shows that:

- Higher values of lag1 increase predictions significantly
- dominantTypeCount also increases predicted crime

This confirms that the model is learning meaningful patterns rather than random relationships.

SHAP improves transparency and helps validate model behaviour.

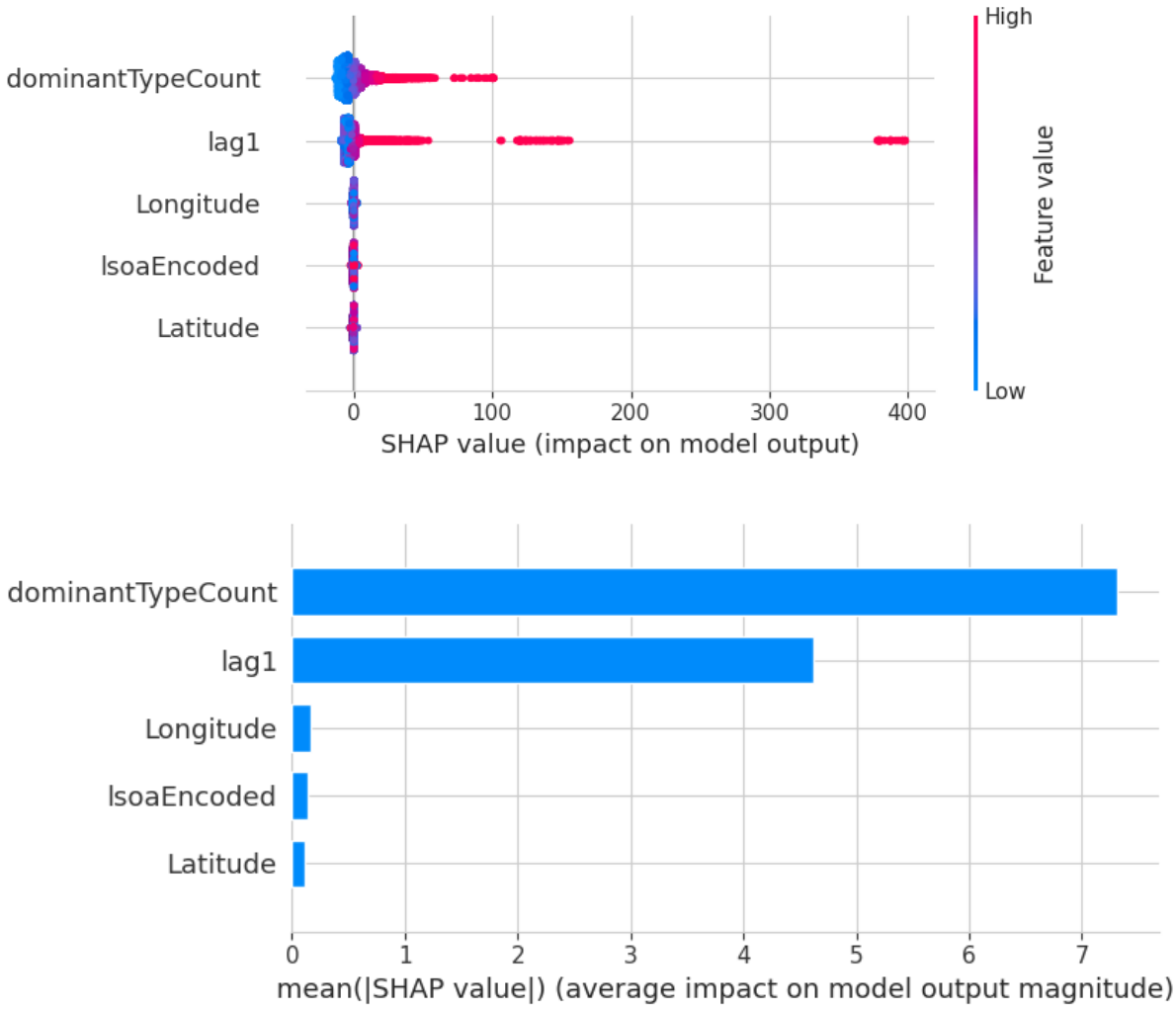

Figure 4.11: SHAP analysis results

The results show that higher values of lag1 lead to higher predicted crime. Similarly, dominantTypeCount increases prediction values. This confirms that past crime behaviour strongly influences future outcomes.

SHAP provides a consistent way to interpret complex models and ensures transparency in predictions.

4.9. Model Validation

Model validation was performed using multiple techniques.

Figure 4.12 shows that predicted values are close to actual values, indicating good accuracy.

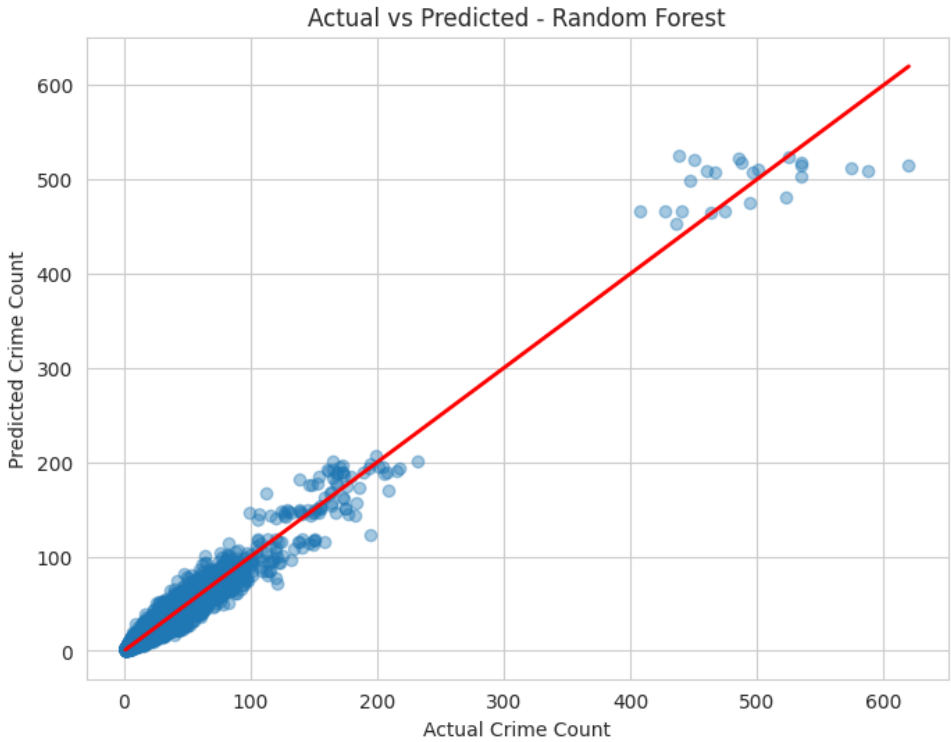

Figure 4.12: Actual vs predicted values

Figure 4.13 shows that most residuals are centred around zero, meaning prediction errors are small and balanced.

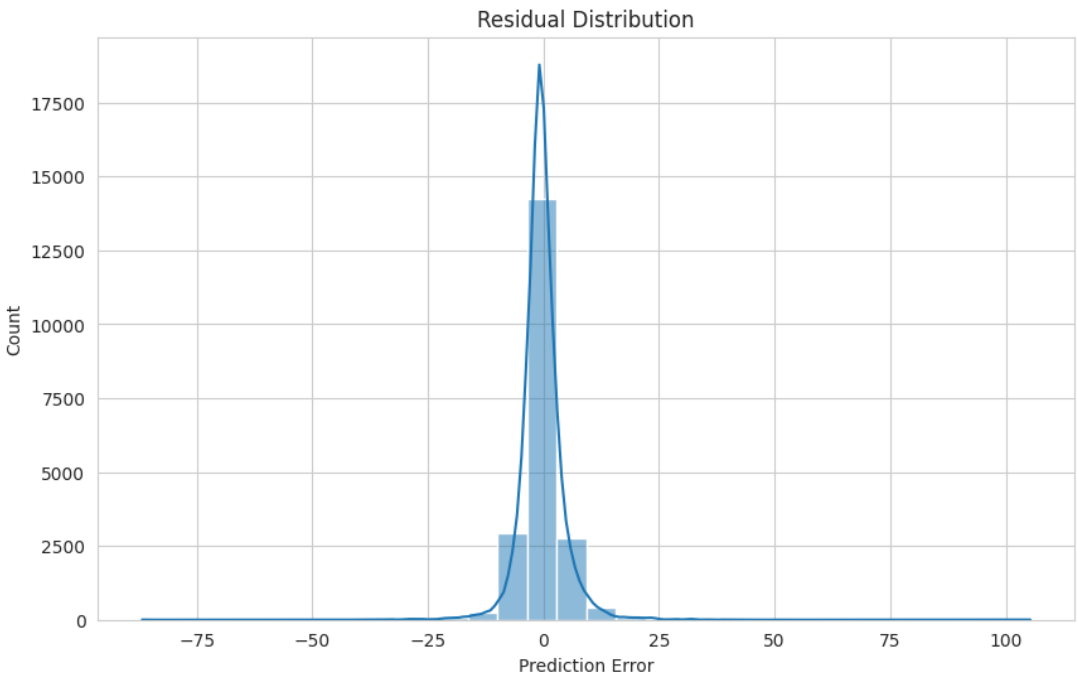

Figure 4.13: Residual distribution

Figure 4.14 shows the learning curve, which indicates that model performance improves with more data but stabilises after a certain point. This suggests the model has learned effectively.

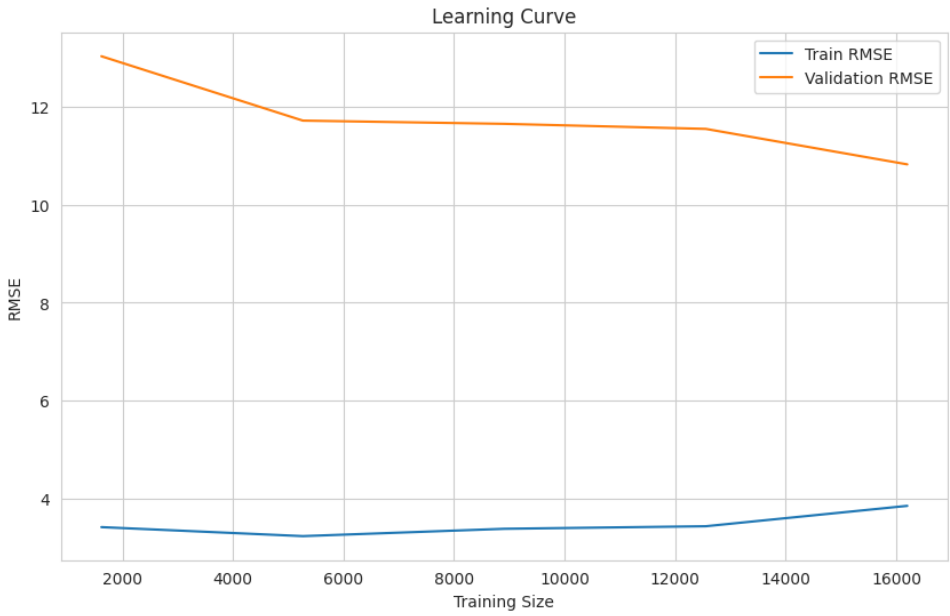

Figure 4.14: Learning curve

The final model achieved an RMSE of **4.62**, improved from earlier values around **4.84**, confirming that tuning enhanced performance.

4.10. Practical Application of the Model

The prediction function demonstrates real-world usability.

Figure 4.16 shows example predictions where the model estimates crime levels and assigns risk categories.


```
{'matchReason': 'Matched by LSOA code: E01032739',
 'resolvedLsoaCode': 'E01032739',
 'lastKnownYear': 2025,
 'lastKnownMonth': 12,
 'predictedYear': 2026,
 'predictedMonth': 1,
 'lastKnownCrimeCount': 535.0,
 'predictedCrimeCount': 490.71,
 'riskBand': 'Very High',
 'dominantCrimeType': 'Violence and sexual offences',
 'latitude': 51.51396745406571,
 'longitude': -0.08723984621673853}
```


Figure 4.15: Example predictions using the model

This allows users to quickly identify high-risk areas and supports decision-making in:

- Policing
- Urban planning
- Public safety

This allows users to quickly identify high-risk areas and supports decision-making in:

- Policing
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5. Conclusion and Recommendations

5.1. Summary of the Study

This study analysed crime data using machine learning to understand patterns and predict future crime levels. The results show that crime varies across time and location, with clear seasonal trends and spatial hotspots.

DBSCAN performed better than KMeans for identifying hotspots, while Random Forest outperformed Linear Regression for prediction. After tuning, the final model achieved an RMSE of **4.62**, showing improved accuracy.

Feature analysis confirmed that past crime (lag1) is the strongest predictor, supported by SHAP interpretation. Overall, combining EDA, clustering, and machine learning provided an effective approach for crime analysis.

5.2. Practical Implications

The results can support real-world decision making. Hotspot detection can help law enforcement allocate resources efficiently, while prediction models enable proactive planning. The developed prediction function can also be used in systems such as dashboards for risk assessment.

5.3. Limitations of the Study

This study has some limitations. The dataset is limited in scope and does not include external factors such as economic or environmental conditions. Aggregation at LSOA level may hide fine details, and the model assumes past patterns will continue in the future.

5.4. Recommendations for Future Work

Future work can include additional features such as population and economic data, use more advanced models, and expand the dataset. Real-time data integration and development of a full decision-support system can further improve practical use.

5.5. Final Remarks

This study demonstrates that machine learning can effectively analyse crime patterns and support data-driven decision making. With further improvements, such approaches can contribute to better public safety and planning.

6. References

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Appendices

Appendix A. Dataset Description

The dataset used in this study was obtained from the UK Police open crime data platform. It contains detailed records of reported crimes, including spatial, temporal, and categorical information. The dataset was originally provided at street level and later aggregated to LSOA-month level for analysis.

Table A.1 shows the key variables included in the dataset along with their description.

Table A:1 Key Variables in the Dataset

Variable	Description
Crime ID	Unique identifier for each crime
Month	Date of the crime
Falls within	Police force area
Latitude	Geographic coordinate
Longitude	Geographic coordinate
Location	Description of crime location
LSOA code	Area identifier
LSOA name	Area name
Crime type	Type of crime
Last outcome category	Outcome of the crime

This table shows that the dataset includes both spatial and temporal features, which are important for analysing crime patterns.

Appendix B. Data Preprocessing and Feature Creation

Data preprocessing was carried out to ensure the dataset was clean and suitable for analysis.

The following steps were applied:

- Removal of missing values in key columns
- Conversion of month column into datetime format
- Extraction of year and month number
- Aggregation of crime counts per LSOA and month
- Creation of new features

Table B.1 shows the features created during preprocessing.

Table B:1: Created Features

Feature	Description
crimeCount	Total number of crimes per LSOA per month
logCrime	Log-transformed crime count
lag1	Previous month crime count
isWinter	Indicates winter months
dominantTypeCount	Count of most frequent crime type

These features help capture temporal patterns and improve model performance.

Appendix C. Feature Set for Modelling

The final set of features used in the machine learning models is shown below.

Table C.1 shows the input features and their purpose.

Table C:1: Model Input Features

Feature	Purpose
lag1	Captures past crime trends
year	Time-based trend
monthNum	Seasonal variation
isWinter	Seasonal effect
Latitude	Spatial location
Longitude	Spatial location
dominantTypeCount	Crime composition
lsoaEncoded	Encoded area identifier

The table shows that both spatial and temporal features are used together for prediction.

Appendix D. Model Configuration

The Random Forest model was selected as the main prediction model due to its strong performance.

Table D.1 shows the parameters used after tuning.

Table D:1: Random Forest Parameters

Parameter	Value
n_estimators	100–300
max_depth	Up to 16
min_samples_split	5
min_samples_leaf	1
max_features	log2

These parameters help control model complexity and improve generalisation.

Appendix E. Evaluation Metrics

Model performance was evaluated using regression metrics.

Table E.1 shows the metrics used in this study.

Table E:1: Evaluation Metrics

Metric	Description
MSE	Measures average squared error
RMSE	Square root of MSE, easier to interpret

RMSE was used as the main evaluation metric because it provides error values in the same unit as crime count.

Appendix F. Aggregated Dataset Sample

The dataset was aggregated at LSOA-month level for modelling.

Table F.1 shows a sample of the aggregated dataset.

Table F:1: Sample Aggregated Dataset

LSOA code	Year	Month	Crime Count	Latitude	Longitude
E01000001	2023	10	6	51.51746	-0.09748
E01000001	2023	11	19	51.51746	-0.09748
E01000001	2023	12	6	51.51746	-0.09748

This table shows how crime data was grouped by area and time for analysis.

Appendix G. Visualisations

This appendix presents the key visual outputs used in the analysis.

Monthly Crime Trend

Figure G.1 shows how crime changes over time. The graph highlights fluctuations and trends across months.

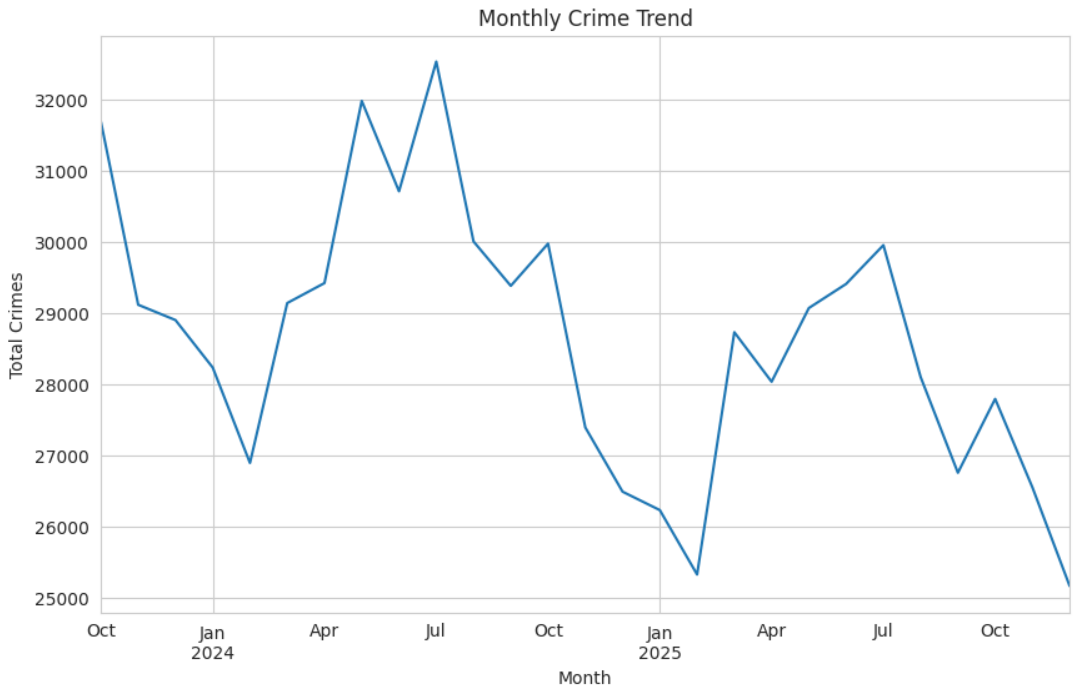

Figure 0.1: Monthly Crime Trend

Seasonal Crime Pattern

Figure G.2 shows how crime varies across different months of the year, helping identify seasonal trends.

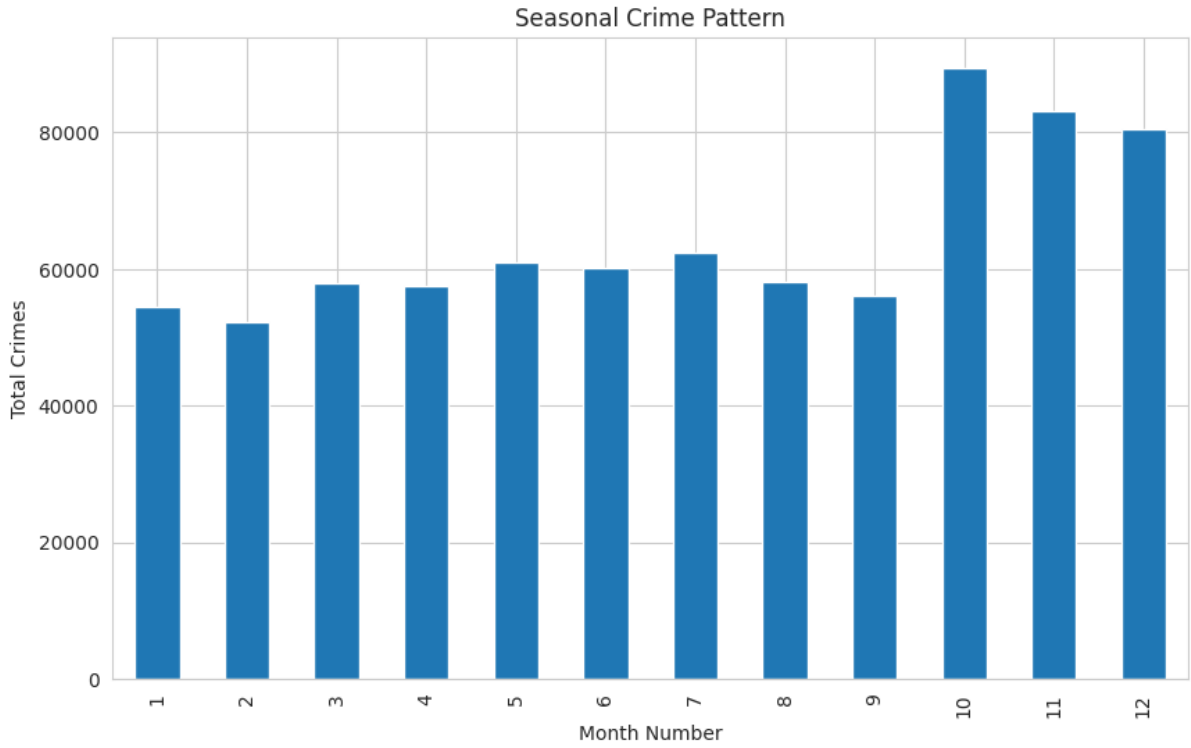

Figure 0.2: Seasonal Crime Pattern

Crime Density Heatmap

Figure G.3 shows the spatial density of crime. Darker areas represent higher crime concentration.

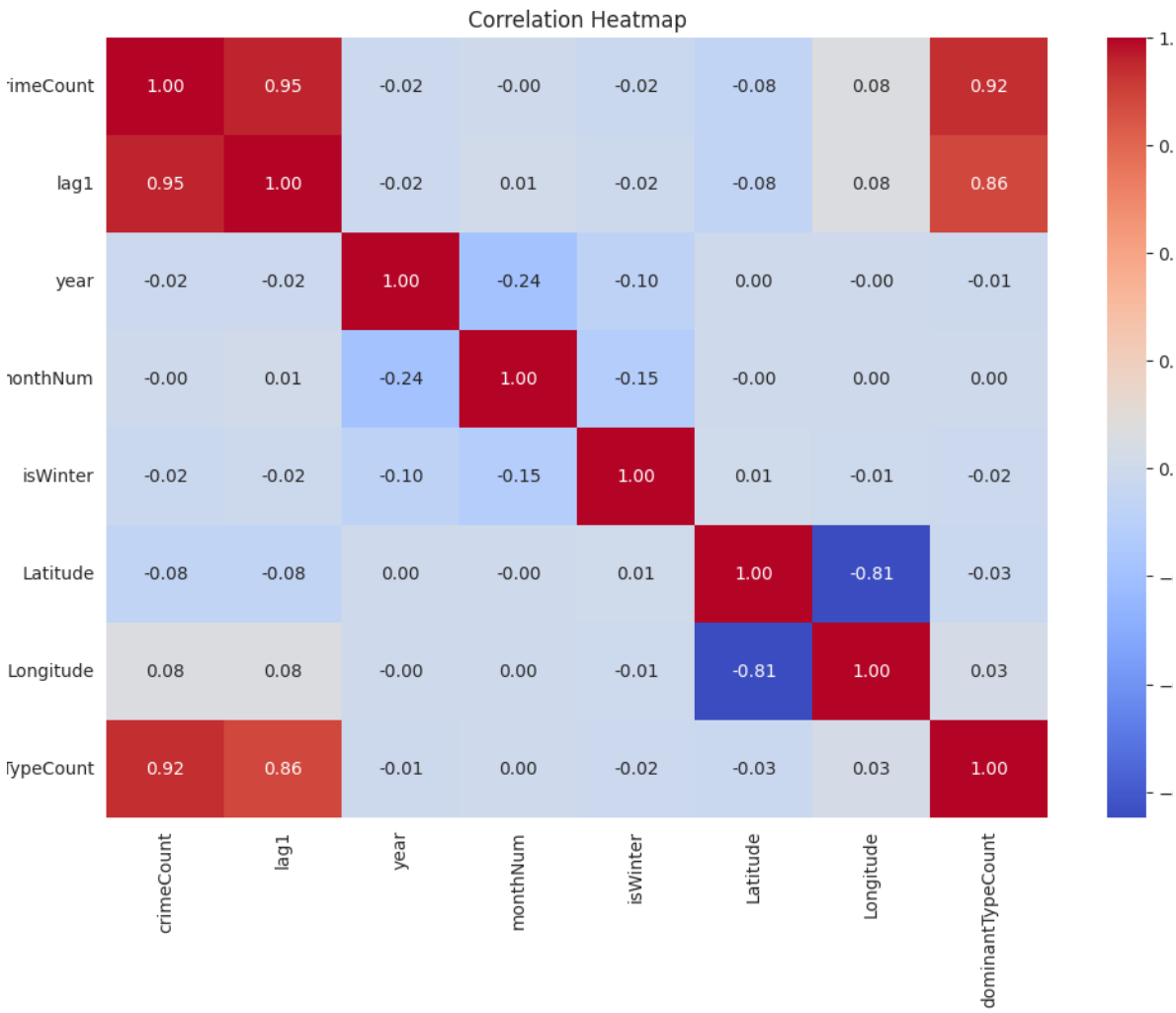

Figure 0.3: Crime Density Heatmap

KMeans Clustering Results

Figure G.4 shows clusters created using KMeans, grouping areas based on similar crime patterns.

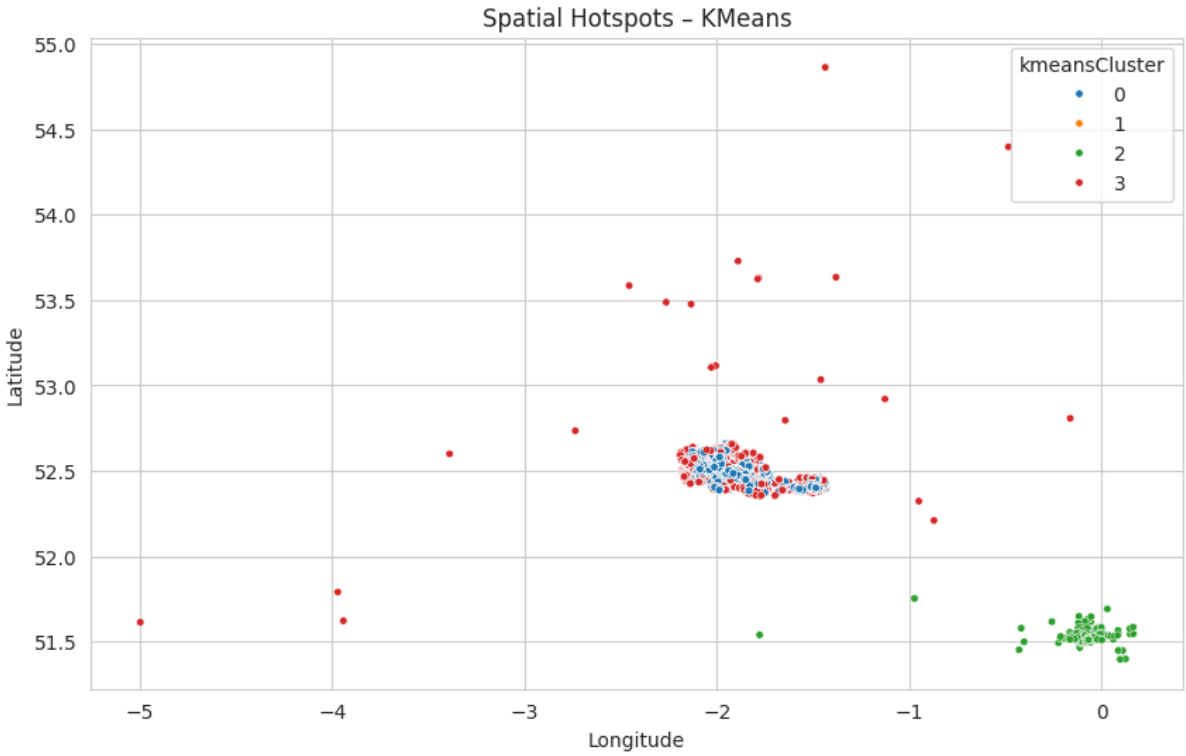

Figure G.4: KMeans Clustering Results
(Insert screenshot)

DBSCAN Clustering Results

Figure G.5 shows clusters identified using DBSCAN. It highlights dense areas and marks outliers.

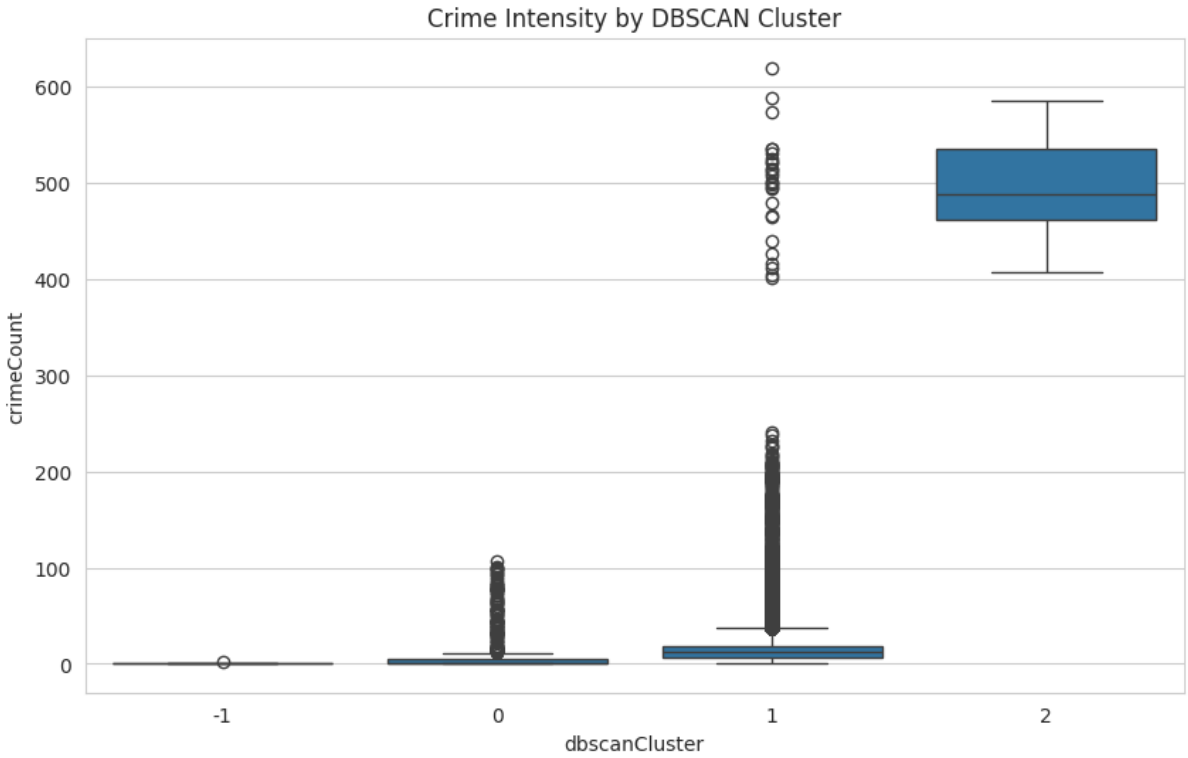

Figure G.5: DBSCAN Clustering Results

Feature Importance Scores

Figure G.6 shows the importance of each feature in predicting crime levels.

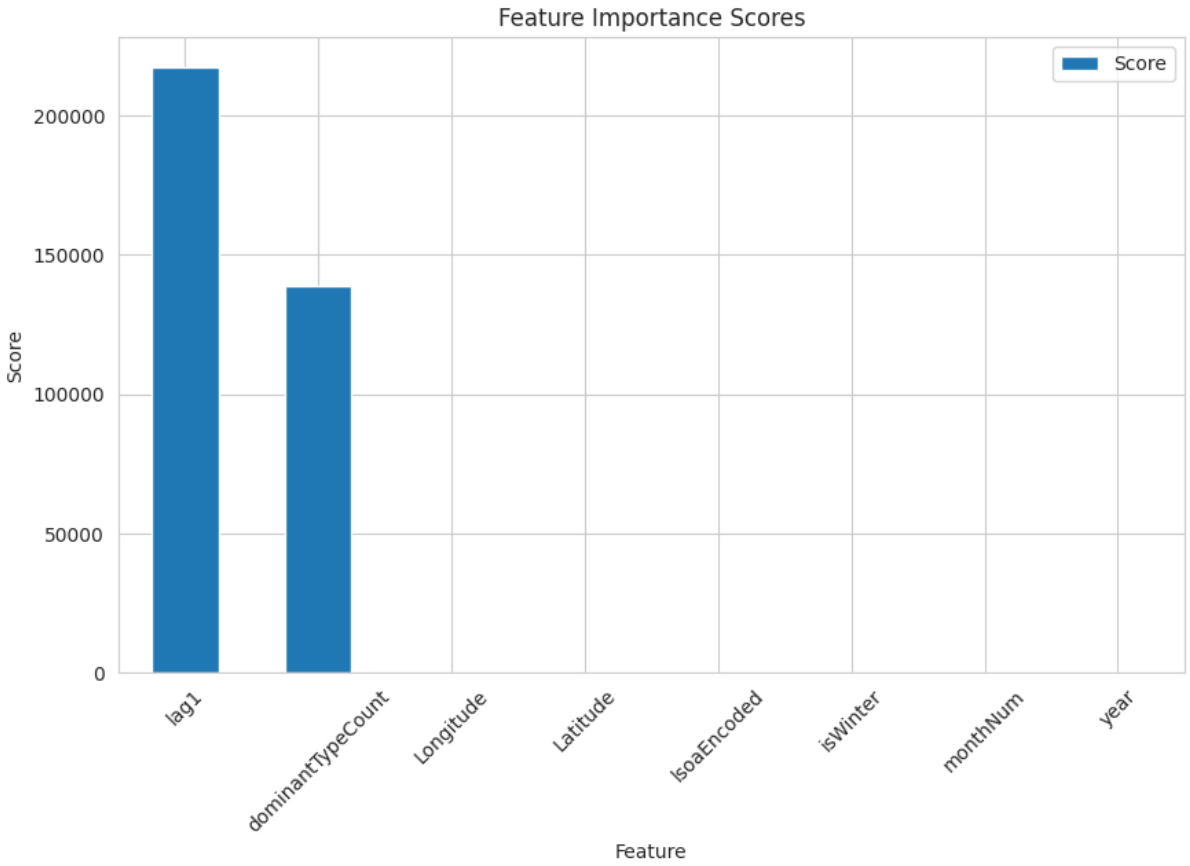

Figure G.6: Feature Importance Scores

Actual vs Predicted Values

Figure G.7 shows the relationship between actual and predicted crime values.

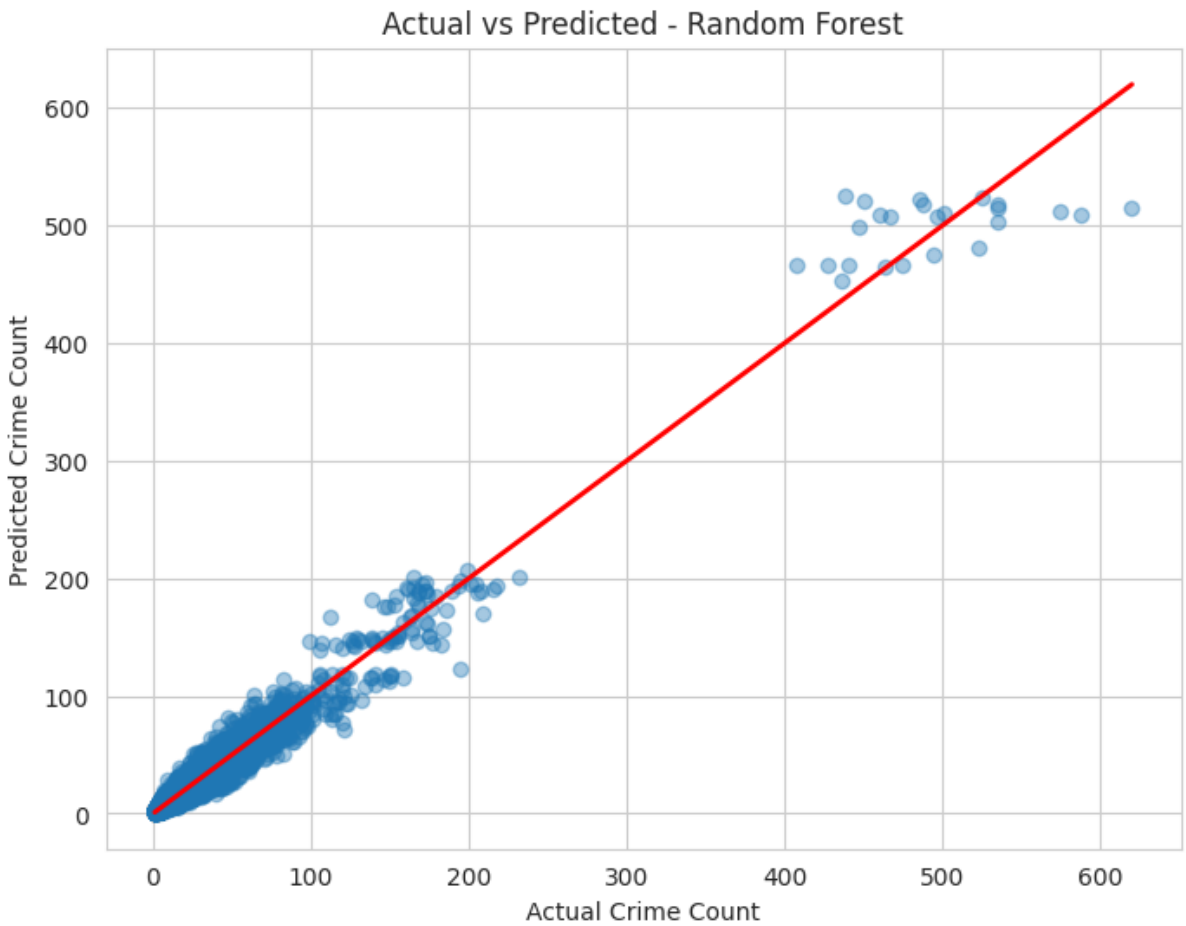

Figure G.7: Actual vs Predicted Values

Residual Distribution

Figure G.8 shows the distribution of prediction errors.

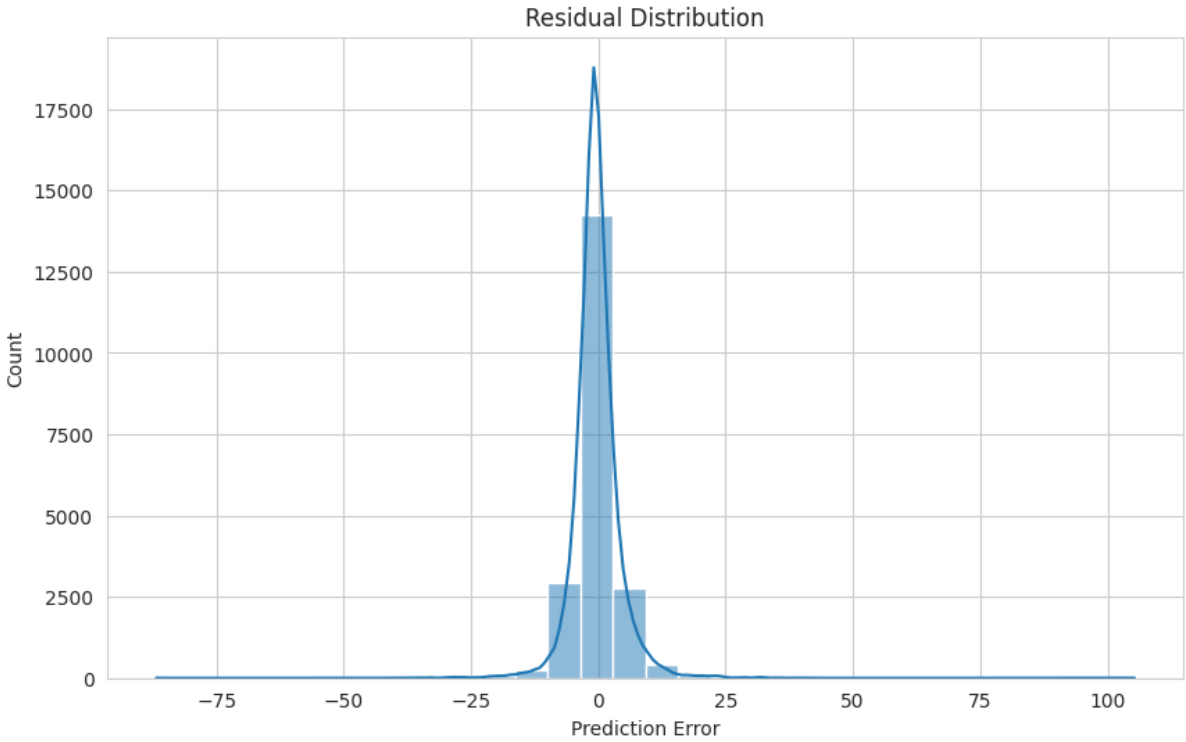

Figure G.8: Residual Distribution

Learning Curve

Figure G.9 shows how model performance improves with more training data.

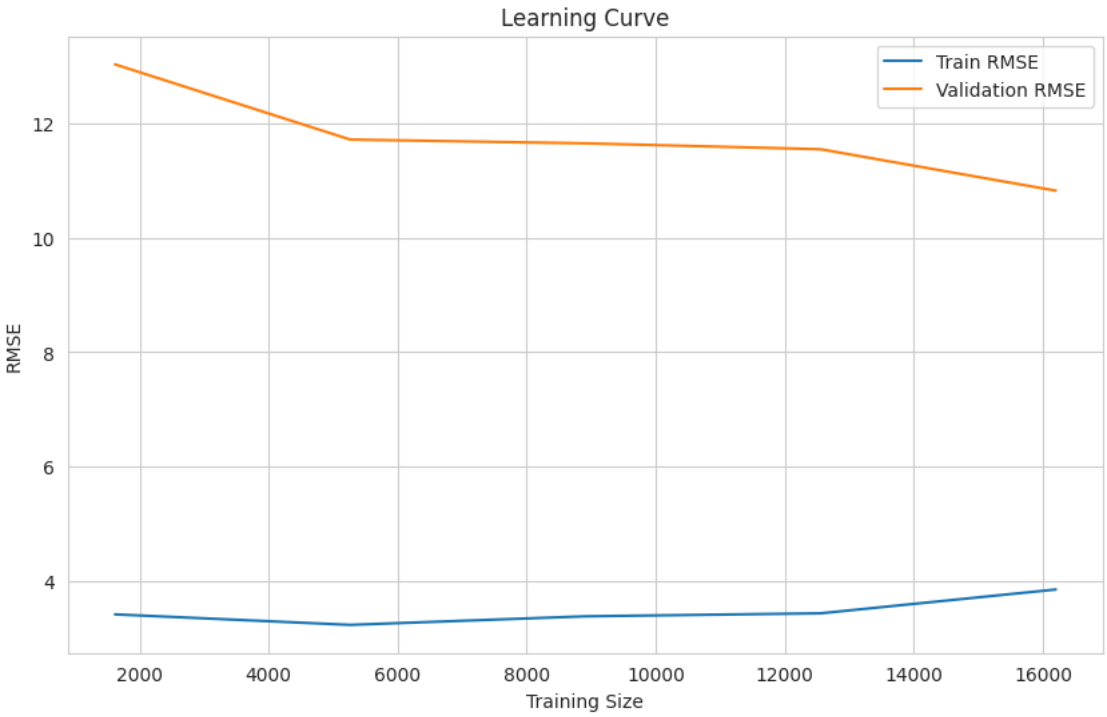

Figure G.9: Learning Curve

Appendix H. Model Performance Results

Model performance was compared across different models.

Table H.1 shows the RMSE values.

Table H:1: Model Performance Comparison

Model	RMSE
Linear Regression	5.35
Random Forest	4.84
Regularised RF	4.85
Tuned RF	4.62

The table shows that the tuned Random Forest model achieved the best performance.

Appendix I. Prediction Function Output

The prediction function was tested using different inputs.

Figure I.1 shows an example output.


```
{'matchReason': 'Matched by nearest coordinates: (51.5139, -0.0872)',
 'resolvedLsoaCode': 'E01032739',
 'lastKnownYear': 2025,
 'lastKnownMonth': 12,
 'predictedYear': 2026,
 'predictedMonth': 1,
 'lastKnownCrimeCount': 535.0,
 'predictedCrimeCount': 490.71,
 'riskBand': 'Very High',
 'dominantCrimeType': 'Violence and sexual offences',
 'latitude': 51.51396745406571,
 'longitude': -0.08723984621673853}
```


Figure 0.4: Example Prediction Output

The output includes:

- Predicted crime count
- Risk level
- Location details

This demonstrates the practical use of the model.